Drishti दृष्टि

Drishti on AWS: the lake, Delta or Iceberg

For platform and infrastructure teams: the lake in S3, the table format chosen per connector, the risk lake, bank scale, the scripts, the keys, the partitions and every optimisation

Delta is the default, Iceberg a full alternative: VaR, intraday and exposure read from either; official results to Delta or Iceberg, on disk or S3 · VaR sharded across servers and the parallel Java loader are built, on either format · only the super-scale scale-out of the whole lake is designed

AWS scripts for both formats tested with --help and --dry-run; Delta on S3, lake publish, the Iceberg engine and official results tested on S3 simulators; the first run in a real AWS account, and the Glue and S3 Tables runs, are still to do

See what your data hasn't told you yet.

ANY DATA. ANY DOMAIN. ONE GRAMMAR.

Ashutosh Sinha · October 2026

Executive summary

	In one line	Slides
The lake lives in S3	Every table's Parquet data and its log or metadata are in S3; EFS keeps only state, caches, the loader's scratch and locks. 200 TB costs about USD 1,400 a month[1] tiered in S3, USD 60,000 in EFS Standard.	4
Delta or Iceberg, per connector	Nothing above the storage plugin knows the format: pages, searches, history and BI give the same 18 answers; VaR and exposure are equal to the cent. Delta is the default; Iceberg is chosen by configuration.	23
One load, one commit a day	JSON Lines land in S3. Delta builds, syncs and publishes one conditional-PUT commit; Iceberg's loader writes S3 and commits itself. A reader sees the old day or the new, never a mixture.	8
The risk lake	VaR and exposure inputs as `list<double>` vectors, computed for any scope; official results append-only for 7 years, to Delta or Iceberg on `s3://`, JSON Lines beside.	45
Bank scale	2.3M positions: about 7 GB of vectors a date[2], 12.4 TB over 7 years (ESTIMATED). VaR sharding and the parallel Java loader are built; the 200K benchmark is PLANNED.	60
Status	Every AWS script dry-run tested; S3 behaviour proven on three simulators; not yet run in a real AWS account; the Glue and S3 Tables runs NOT RUN.	54
Three machines	The smallest highly available Drishti: three m7i.8xlarge, the lake in S3, RDS Multi-AZ, an ALB: about USD 5,400–6,100 a month[3] on demand in Canada (Central), USD 3,500–4,200 with a 3-year Savings Plan (ESTIMATED).	94
To decide	Hot files on NVMe; the first run in a development account; the format per domain; the Iceberg catalog; the fixed gossip port; compaction on S3.	106

[1] tier split × S3 list prices = USD 1,381; EFS: 200,000 GB × USD 0.30 [2] 2.3M × 509 × 8 B × 0.75 Parquet ratio; × 1,764 dates [3] three machines on AWS: the cost slide's sum, Canada (Central), on demand; RDS an ASSUMPTION Full derivations: slides 133–136.

What this deck covers

	Chapter	What it holds	Slides
01	Why S3	the one rule: the lake lives in S3, whatever the format; S3 against EFS at 200 TB	4
02	How it works	the flow, JSON Lines in, prefixes, partitions, files, one commit a day, safe commits, caches	7
03	Delta or Iceberg	where each plugs in, side by side, Iceberg as built, the decision guide, converting, running both	23
04	The risk lake	vectors, intraday and Kafka batches, official results to Delta or Iceberg, exposure paths	45
05	Evidence	status and tests, S3 simulators, Delta = Iceberg, bank scale, the loader, sharding, the demo lake	53
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Why S3

01



The one rule: the lake lives in Amazon S3

Every table's Parquet data files AND its log (Delta `_delta_log/`) or metadata (Iceberg `metadata/`) are in S3. Nothing of the lake is on EFS. Delta or Iceberg is chosen per connector.

EFS holds only what a node must find again after a restart: configuration, users and tokens, Drishti*net state, the content store's filesystem backend if chosen, and the caches that should be warm after a restart.

- **S3 (the lake):** Delta: `s3://<org>-drishti-<env>-lake/lake/<domain>/<table>/` with `business_date=…/PRODUCT_TYPE=…/part-….snappy.parquet` and `_delta_log/`; Iceberg: `…-lake/iceberg/<domain>/<table>/` with `metadata/` and `data/business_date=…/`
- **Domains:** trading, reference, legal, collateral, credit, market, macro, news, civic, **risk** (VaR and exposure inputs), **risk-results** (official results, append-only), us-equities, gleif, funds
- **S3 (landing):** the sources' dumps as JSON Lines, `<date>/<source>/*.jsonl.gz`, kept 30 days (`LANDING_KEEP_DAYS`)
- **S3 (logs):** access logs, SSE-S3, expire after 400 days
- **EFS (state):** `/drishti/<env>/<node>/opt/drishti/{data,config,sutras}`, one access point per node per path, identical path inside the container
- **EFS (loader scratch):** one day of one table for the minutes of a load, then removed; the per-table lock files
- **Never on EFS:** lake data, `_delta_log/`, `metadata/` (EFS is a staging place only: the loader's scratch, the Kafka stream's windows)

Why S3 and not EFS: the comparison

	Amazon S3	Amazon EFS
Cost of 200 TB (list prices, one Region)	about **USD 1,400 a month**[1] tiered by business date; USD 4,600 all in Standard[2]	**USD 60,000 a month**[3] in EFS Standard; USD 3,200 in Infrequent Access[4] plus a charge on every byte read
Request rate	at least 3,500 writes and 5,500 reads a second per prefix; every `business_date=…/` folder is a prefix	one file system's limits, shared by every node, loader and maintenance job
Latency of a single read	10–30 ms to the first byte in-Region; Drishti reads one row group	about 1 ms for hot data in Standard; tens of ms in IA
Durability	eleven nines, ≥ 3 AZs, versioning, Cross-Region Replication	Regional (three zones); backup by AWS Backup
Tiers	Standard → Standard-IA → Glacier Instant Retrieval (still ms), by date	Standard → IA → Archive by last access only
Other readers	Athena, Trino, Spark, EMR, Glue read the same Delta tables; Snowflake too for Iceberg (through a catalog)	only something that mounts the file system
Writers	conditional writes make concurrent commits safe (the native Iceberg engine probes for them)	a hard link on a POSIX file system (what Drishti uses on local disk)
Iceberg only: file paths in metadata	`s3://bucket/…` is the same for every task; Cross-Region copies read with `relocate`	the path depends on the mount; a moved folder is a moved table

At 200 TB+ EFS costs 10–40×[5] more and shares one throughput budget, whatever the table format. Drishti reads one day's files, one row group of one file, by byte ranges. **The lake is in S3; EFS keeps the small, hot, per-node state.**

[1] tier split × S3 list prices = USD 1,381 [2] 200 TB × USD 23 a TB-month [3] 200,000 GB × USD 0.30 a GB-month (EFS Standard) [4] the doc's figure (implies USD 0.016 a GB-month; to be confirmed) [5] 60,000 ÷ 1,381 ≈ 43×; 60,000 ÷ 4,600 ≈ 13× Full derivations: slides 133–136.

How it works

02

The picture: the whole flow on one page

Sources

- ENDUR, CALYPSO, MUREX, reference, risk batch
- JSON Lines `.jsonl(.gz)`; CSV deprecated

S3 · <org>-drishti-<env>-landing

- `2026-10-20/endur/*.jsonl.gz`, …/calypso/, …/murex/
- kept `LANDING_KEEP_DAYS` (30), then expired

EventBridge Scheduler

- 22:30 New York, Monday to Friday
- runs the loader task; nightly maintain, daily report

ECS task · loader · load-day.sh, per table, under the table's lock (EFS)

- **Delta:** 1 sync landing → EFS scratch 2 `lake build` → scratch table (sorted by id, 1 MB row groups) 3 sync data files → S3 4 `lake publish` ONE commit
- **Iceberg:** `IcebergLoader` reads the dumps, sorts each day by id, writes Parquet + manifests to S3, commits `vN+1.metadata.json` (conditional PUT)

S3 · <org>-drishti-<env>-lake (THE LAKE)

- `lake/…/_delta_log/` · `iceberg/…/metadata/` + `business_date=…` Parquet
- domains: trading … market, **risk**, **risk-results**, public data
- versioning · SSE-KMS · VPC endpoint only · tiers by business date

ECS services drishti-a, -b, -c (server + console)

- `delta` and `iceberg` connectors, native engines, ranged GETs
- VaR and exposure engines read `risk/`; caches in heap and on disk
- Drishti*net: gossip port + HTTP/2 between task IPs

ECS task · maintenance (nightly)

- Delta: checkpoint, vacuum · Iceberg: expiry, orphans · retention, 7-year floor

EFS (Regional, encrypted, Elastic)

- state, caches, loader scratch, locks; NO lake data

The picture, in words: each box

Box	What it is	Key facts
Sources	ENDUR, CALYPSO, MUREX, reference systems, the risk batch	JSON Lines, gzipped or not; one folder per date and source; JSON arrays still read; CSV input deprecated
Landing bucket	`<org>-drishti-<env>-landing`	`<date>/<source>/…`; dumps expire after 30 days (`LANDING_KEEP_DAYS`); same protections as the lake
Scheduler	EventBridge Scheduler	`LOAD_SCHEDULE` 22:30 in `LOAD_SCHEDULE_ZONE` (New York), Mon–Fri; also -maintain and -report
Loader task	ECS task `drishti-<env>-loader`	**Delta:** sync → build → sync → publish → metrics, per table under its EFS lock; scratch always removed. **Iceberg:** `IcebergLoader` (the server's exec jar): sorts each day by id, files of 250,000 rows, one overwrite of the day's partition, no publish step
The lake	`<org>-drishti-<env>-lake`	Delta: Parquet + `_delta_log` under `lake/`; Iceberg: `metadata/` + `data/business_date=…/` under `iceberg/`, files named by absolute paths; versioning, SSE-KMS, VPC endpoint only
Server nodes	ECS services `drishti-a/-b/-c`	server + console per task; `delta` and `iceberg` connectors, native engines, one row group per entity; Iceberg refreshes every 10 s; VaR and exposure engines; Drishti*net
Maintenance task	ECS task, nightly (`MAINTAIN_SCHEDULE`)	`lake maintain` per table under its lock (Iceberg: `IcebergMaintenance`, expiry, orphan files ≥ 72 h); risk domains never below 7 years; then setup-lifecycle.sh
Catalog (Iceberg, optional)	a REST catalog: Glue's Iceberg endpoint, Polaris, Snowflake Open Catalog	commits by the catalog's requirements (409 retried); vended credentials; path-based tables need none. On AWS: NOT RUN
EFS	Regional, encrypted, Elastic throughput	`/drishti/<env>/<node>/opt/drishti/{data,config,sutras}`; loader scratch, spill and locks; never lake data, `_delta_log` or `metadata/`

Generate JSON Lines, then load it into Delta

Source or generator
- writes JSON Lines
- `trades-<date>.jsonl.gz`

→

Landing / build folder
- `<date>/<source>/*.jsonl.gz`
- a restatement: a second file

→

lake build · load_rows.py
- promoted and vector columns
- sorted by id; one commit a date

→

Delta table
- `lake/<domain>/<table>/`
- read by the Delta connector

Form	One line is	Written by → read by
Document	one entity as its source holds it: `{"TRADE_ID": "1000015", "LEGS": [...]}`	make_trades, bulk_trades, pack samples → `lake build`, the file connector
Loader row	`{"domain", "kind", "id", "date", "doc", "columns"}`	`make_data.py --jsonl` → PostgreSQL, Cassandra, Iceberg … loaders
Lake row	`{"domain", "kind", "table", "id", "date", "doc"}`	`make_data.py --lake`, generators → `load_rows.py`
Staged lake row	`{"id", "doc", "columns"}`	lake build's staging, official-results publishers → Java `LakeWriter`

The rule: one JSON object per line, UTF-8, `.jsonl` or `.jsonl.gz`. Generating and loading are separate steps; a loader never rewrites a pack's samples. The lake built from JSONL is byte-for-byte the lake built before.

CSV in is deprecated; JSON Lines out beside every format

Coming in

- **JSON Lines first**; JSON arrays and per-entity `.json` still read
- **CSV input deprecated**: the file connector still serves feed files, warns once per file, names them in Admin → Health
- `drishti.py convert trades.csv` → `trades.jsonl` (also JSON arrays, gzip, folders)

Going out

- every export and download offers JSONL **beside** CSV, Excel, PDF, PNG
- server: `search/jsonl`, SQL `format=jsonl`, BI runs, statements, VaR and exposure captures
- console: one ↓ per panel, CSV or JSON Lines; CLI `--jsonl`

```
drishti.py convert trades.csv                                    # trades.jsonl, a line per row
drishti.py convert data/feeds/fixing/SOFR-HISTORY.csv --rows-document --out data/feeds/fixing.jsonl
drishti.py convert data/feeds/settlement                         # settlement.jsonl, the file name as "id"
```

Not data, so not JSONL: Delta and Iceberg Parquet, the BI store, the VaR and exposure hot files, Arrow between the server and Python, manifests. **The guard:** two drill tests check that every generator writes JSON Lines and every export entry point offers it.

Prefixes and tables

Choice	Why
One table per source system and kind (`trade_endur`, `_calypso`, `_murex`, read as a union)	each source loads, restates and fails on its own; each log a third the size
Domain as the first folder	matches the connector's domain; IAM and lifecycle rules can name a domain
`risk/` one table per input kind; `risk-results/` written only by the publishers	engines read `risk/`; no engine reads `risk-results/` (a guard test)
Tables by year (`trade_endur/2026/`)	DESIGNED (S2): until then one table holds every year

```
s3://acme-drishti-prod-lake/
└── lake/                    DRISHTI_DELTA_ROOT
    ├── trading/
    │   ├── trade_endur/   _delta_log/
    │   │   └── business_date=2026-10-20/
    │   │       └── PRODUCT_TYPE=IRS_FIXFLOAT/part-….parquet
    │   ├── trade_calypso/ …
    │   └── trade_murex/ …
    ├── reference/ counterparty/ market/ …
    ├── risk/                the engines' inputs
    │   ├── pnl-vector/  scenario-pv/  fx-scenario/
    │   ├── scenario-set/  risk-hierarchy/  var-limit/
    │   └── portfolio/  simulation-run/  exposure-path/
    ├── risk-results/        append-only, never read back
    │   ├── var-official/  exposure-official/ …
    │   └── jsonl/<table>/<date>/
    └── us-equities/ gleif/ funds/   public data
```

An Iceberg table in S3: metadata/ beside data/


```
s3://acme-drishti-prod-lake/
├── lake/                       Delta tables (DRISHTI_DELTA_ROOT)
└── iceberg/                    DRISHTI_ICEBERG_ROOT = s3://…/iceberg
    └── trading/                the domain (a REST catalog's namespace)
        ├── trade_calypso/      one table per source system and kind
        │   ├── metadata/
        │   │   ├── v12.metadata.json   schema, spec, sort order,
        │   │   │                       properties, snapshots
        │   │   ├── version-hint.text   path-based tables only
        │   │   ├── snap-…-….avro       a manifest list per snapshot
        │   │   └── …-m0.avro           manifests: files, partitions,
        │   │                           column bounds
        │   └── data/
        │       ├── business_date=2026-10-19/
        │       └── business_date=2026-10-20/
        │           ├── 2026-10-20-00000-<uuid>.parquet  ids A… – C…
        │           └── 2026-10-20-00001-<uuid>.parquet  ids C… – Z…
        └── trade_murex/ …
```

Choice	Why
One table per source system and kind	each source loads, restates and fails on its own (`table.trade: trade_calypso`)
Domain as the first folder	the connector's `domain`; IAM and lifecycle can name it
`metadata/` beside `data/`	the whole table under one prefix: copy, replicate, grant as a unit
Absolute paths in metadata	a moved table needs `relocate` (native engine)

Partition keys and request-rate-friendly prefixes

Key	Delta on AWS	Iceberg	Why
`business_date`	always the first key	the only partition (identity)	every interactive read names a date; a load and a restatement replace one day; retention removes whole days; tiers by the folder's date
`PRODUCT_TYPE` **(trades)**	second key (`--partition-by`)	not a partition	spreads a Delta day over several prefixes, keeps each partition sortable in one batch, lets a source restate one product type; Iceberg sorts a whole day externally
nothing finer	two keys for trades, one for the rest	`business_date` only	each partition column multiplies the folders and small files of a day
risk tables	`business_date` only	`business_date` only	the engine reads a whole date's vectors in one streamed pass; it sorts into hierarchy order in memory
hidden transforms	—	read for any table (`day(ts)`, `bucket(id)`)	other engines' tables, served by `iceberg-external`

- S3 scales each prefix to ≥ 3,500 writes and 5,500 reads a second and splits busy prefixes by itself; a day's reads spread over its product-type folders, and over three tables for trades
- Drishti's writers name the folders, so the spread comes from the partition keys; Iceberg's hashed object-storage layout is not used (evaluate only if 503 Slow Down appears: PROPOSED)
- A 503 Slow Down is retried with back-off and counted in Health; alarm `drishti-<env>-lake-s3-5xx` fires on a sustained rate

What each read needs from the table

Need	What it reads	So the table needs
One entity (`TRD MX-20000017`)	the day's id map, then one row group of one file	rows sorted by id per file; small row groups
History (`TRD MX-20000017 30`)	the days whose id map holds it, one read per day, eight at once	`business_date` as the partition; nothing else
Type-ahead	the newest day's ids (and `index.days`), plus promoted name columns	`id`; `layout.<kind>.type-ahead` for names
Search, pick lists, derived kinds, impact, links	the day's column set: id and every promoted column, never `doc`	every field a search filters, sorts or shows promoted
A date's vectors (VaR, exposure)	one vector column and a few text columns, streamed file by file	`layout.<kind>.vectors`; whole dates, no deletion vectors
File and row-group skipping	Delta: per-file min/max in the log; Iceberg: manifest bounds per file; both: footer statistics per row group	Delta: `delta.dataSkippingStatsColumns` = id, business_date, promoted; Iceberg: metrics full on `id`, on promoted, none on `doc`

The numbers: Delta and Iceberg

Setting	Delta on AWS	Iceberg	Why
Rows per file	`layout.<kind>.file-rows` (250,000)	`--file-rows` 250,000 (≈ 175 MB, estimate[1])	few files per day, each large enough for S3
Row group	1 MB (`LAKE_ROW_GROUP_KB=1024`; writer default 256 KB)	1 MB **compressed** (`--row-group-mb`): 300–700 trades	a single read fetches one row group: 1 MB balances bytes against 10–30 ms GETs
Compression	Snappy (the native engine also reads ZSTD, GZIP, LZ4_RAW)	zstd (about 9× on documents)	the writers' choices
Statistics	the log: id, business_date, promoted	manifests: `id` full, promoted, none on `doc`	file and row-group skipping without documents in the metadata
Sort	by id within a batch (`LAKE_BATCH_ROWS` 250,000)	by id over the whole day (external sort, spill)	no partition-size limit on Iceberg
Commit unit	one per table per day (`lake publish`)	one per day of ≥ 100,000 rows; small days together	fewer commits for small tables
Checkpoint, metadata	a checkpoint every 10 commits and after every maintenance run	50 metadata versions kept, older deleted after a commit	a reader replays a short log, or reads one small pointer
History kept	log 30 days (`delta.logRetentionDuration`); removed files 7 days (`LAKE_VACUUM_HOURS` 168)	snapshots 168 h (`--expire-hours`)	readers in flight and *known at* reach back a week; **7 years for risk** on both

[1] 250,000 rows × 0.7 KB a trade (zstd, measured, ICEBERG_CONNECTOR.md §3) = 175 MB

Sorted by id: Delta in batches, Iceberg by the day

- A file holds a contiguous id range and each row group a smaller one: a read of one id skips every other row group by its statistics (Iceberg's manifests keep each file's min and max `id`)
- A table written elsewhere (Spark): `repartitionByRange(…, "id").sortWithinPartitions("id")`; for Iceberg also `WRITE ORDERED BY id`

Delta: `lake build` sorts a partition in one batch

- each partition (date × product type) that fits one batch (`LAKE_BATCH_ROWS`, 250,000) is sorted; a larger one is written in the dumps' order and load-day.sh warns
- **remedy 1:** ask the source to export sorted by id: the JSON Lines order becomes the file's order, no memory cost (`bulk_trades.py` does so)
- **remedy 2:** partition finer (`PRODUCT_TYPE`, or a book or desk column)
- **remedy 3:** raise `LAKE_BATCH_ROWS` with the loader's memory: rows × document size × 3; 250,000 trades of 10 KB ≈ 8 GB[1]; the loader has 30 GB

Iceberg: the loader sorts the whole day

- read JSON Lines from a file or stdin; dates after tomorrow refused
- sort per day in memory up to `--buffer-mb`, spilled runs beyond: the spill directory needs about the JSON's size; **no batch limit**
- merge the runs in id order into files of 250,000 rows
- ≈ 40 s for a day of 1,000,000 trades[2] on one thread, 4 files, 0.70–0.75 GB (ESTIMATED from a trial); 17,910 sample rows into 46 tables in 8 s, 30,000 trade-days in 5 s (MEASURED)

DESIGNED (SUPER_SCALE S1): range maps from the log's per-file id ranges instead of id maps, and Parquet Bloom filters on id.

[1] 250,000 rows × 10 KB a document × 3 (the doc's memory rule, AWS_DELTA_LAKE.md 7.2) = 7.5 GB ≈ 8 GB [2] one writer thread on a trial of 1,000,000 trades (ICEBERG_CONNECTOR.md §6); files 1,000,000 × 0.7 KB ≈ 0.70 GB in 4 files of 250,000 rows

One commit per day per table

Delta (`load-day.sh`)

lake build
- writes the day into a FRESH scratch table on EFS: every file there is live

→

aws s3 sync
- copies the data files to S3: new UUID names, nothing in S3 is overwritten

→

lake publish
- ONE commit that adds the new files and removes the day's previous files

Iceberg (`IcebergLoader`)

read
- the day's JSON Lines

→

sort
- per day, spilled runs beyond `--buffer-mb`

→

merge
- runs in id order; files of 250,000 rows

→

commit
- ONE overwrite of the day's partition

- A reader sees **the old day or the new day, never a mixture**; the previous version stays readable (*known at*) until vacuum (`LAKE_VACUUM_HOURS`, 168 h) or snapshot expiry (`--expire-hours`, 168); 7 years for risk
- Delta: about **1,500 commits a year per table**[1] including maintenance; `lake publish` checks every file is in S3 (on the store it commits to) and refuses a change of columns
- Iceberg: a day of ≥ 100,000 rows is committed as soon as written, smaller days together; loading a day again replaces it

`lake build` and `lake demo` make one commit per business date (not per partition): sixty days of `trade_endur` went from 300 commits replaying the whole log (98 s) to 60. Intraday risk batches are the one exception: an appended commit per batch (slide 48).

[1] AWS_DELTA_LAKE.md 12.1's estimate: 252 load commits plus the nightly maintenance commits (retention, checkpoint, vacuum; Tue–Sat); not broken down in the doc

Why concurrent commits on S3 are safe, in both formats

Writer	How it commits
`lake publish`, `lake maintain` (Python deltalake 1.x)	conditional PUT of the commit file (`If-None-Match: *`): the second writer gets 412, re-reads the log and retries (20); no DynamoDB lock; unsafe rename refused
Drishti's Java Delta writer (`lake build`, Kafka windows)	local disk and EFS: a hard link of a finished commit file
The official-results publisher	on `s3://` each Delta commit file is an exclusive create (`If-None-Match: *`): two servers both land, the second on the next version
Drishti's Iceberg writer, S3 path-based	conditional PUT of `vN+1.metadata.json`, probed at the first commit; a store that ignores it makes the table **single-writer**; then `version-hint.text`
Drishti's Iceberg writer, local disk or EFS	the finished `vN+1.metadata.json` created under its final name by a hard link (fails if taken): 4 writers at once lose nothing
Iceberg behind a REST catalog (Glue, Polaris)	the catalog's commit with requirements (`assert-table-uuid`, `assert-ref-snapshot-id`, ...); a **409** is re-read, re-applied, retried
The content store's s3 backend	`If-None-Match: *` to create, `If-Match: <ETag>` to update; the loser gets `409 DRS-5057`

Two Iceberg writers on one table: one wins each version; the other re-applies its change and retries (10). A re-applied rewrite of a day that changed meanwhile is refused; an append always re-applies; files of a commit that finally failed are deleted at once.

Two layers of safety. `check-conditional-writes.sh` proves the bucket refuses a second create; on top, the scripts take a lock per table on EFS (flock, NFSv4 byte-range locks, released if a task dies). **Tested:** three `lake publish` runs at once on moto, one winner per version; racing Iceberg writers on MinIO. No DynamoDB lock table in either format.

The content store: cluster state on S3 without a shared disk

One interface, four backends (config/store/store.yaml)

- get · compare-and-set put · list · delete · changes

filesystem

- default; a shared mount (EFS) on a cluster

sqlite

- single node

postgres

- one SQL table

s3

- conditional PUT; ETag versions

- **s3:** `<prefix>/items/<key>`; create `If-None-Match: *`; update `If-Match: <ETag>`; a lost race is `409 DRS-5057`
- Two nodes saving one item at once: one success, one `409`; an unreachable store `503 DRS-5059`
- A dead node's BI run is adopted by exactly one node, by a compare-and-set lease

What it holds on a cluster

- Designs; BI workspaces, items, versions, runs, schedules
- data-load overrides, connector edits, collaboration
- VaR and exposure captures and official records (append-only; row lists as JSONL)

On ECS

`s3` or `postgres` need no shared disk; `filesystem` needs EFS at the same path on every node. The s3 bucket needs the conditional-write check of step 4.

What each server caches, Delta and Iceberg

Cache	Where, bounded by	Keyed by (Delta · Iceberg)	After a restart
file list / layout of each table	heap: Delta from the last checkpoint; Iceberg per snapshot (`layout-cache` 64)	version · snapshot	re-read from the checkpoint or the metadata
Parquet footers	heap (native engines), 50,000 row groups, shared	the file (never changes)	re-read on first use (one ranged GET)
manifests, delete files (Iceberg)	heap: 2,000,000 manifest entries; 5,000,000 delete rows	the file	re-read on first use
id maps of days	heap (20 % shared) and disk (`id-map-disk.dir`, memory-mapped)	table version · **the day's fingerprint**	memory-mapped from disk: no scan
column sets of days	heap (`columns-cache-mb`; 1,024 on Iceberg)	as id maps	re-read (newest day in the background)
documents	heap and RocksDB on disk (`doc-disk-cache.dir`, 1 GB, 12 h)	as id maps	warm from disk
history answers	heap (`drishti.history.cache-mb`, 256 MB)	the kind's data version	rebuilt on ask
VaR hot files	`DRISHTI_VAR_HOT_DIR`: fast local disk; 2 dates open, 60 on disk	the date's inputs	rebuilt from the lake (deleted at shutdown by default)
exposure hot files	`DRISHTI_EXPOSURE_HOT_DIR`: local disk; dates within retention	the date's run	rebuilt from the lake

Iceberg's edge: a commit of a new day leaves every other day's caches in place; on Delta a commit moves the table version, so untouched days reload (keeping them: PROPOSED). Disk caches and hot files default under `<data dir>`, on ECS on EFS.

EFS or local NVMe for the disk caches and hot files

	EFS (default)	Local NVMe (instance store)
Latency	≈ 1 ms an operation (NFS); page cache serves repeated reads	tens of microseconds
After a task moves host or zone	warm: same folders	empty: rebuilt as users read
Launch type	Fargate or EC2	EC2 only (m6id, r6id, i4i)
Cost	EFS Standard ≈ USD 0.30/GB-month[1] + Elastic charges	included in the instance
Identical-path rule	satisfied	not: the cache path is on the host
Memory-mapped VaR and exposure hot files	works, but each page fault is an NFS read	what the guides ask for: fast local disk

```
# config/connectors/trade-endur.yaml
id-map-disk.dir:
  /opt/drishti/nvme/cache/trade-endur/idmaps
doc-disk-cache.dir:
  /opt/drishti/nvme/cache/trade-endur/docs
# the server task
DRISHTI_VAR_HOT_DIR=/opt/drishti/nvme/var/hot
DRISHTI_EXPOSURE_HOT_DIR=/opt/drishti/nvme/ccr/hot
```

Recommendation

Start the connector caches on EFS. Put the VaR and exposure hot files on NVMe from the start (`register.sh --nodes --cache nvme`, EC2): they are rebuildable and memory-mapped. The NVMe exception is the product owner's decision.

[1] EFS Standard list price, one Region, as quoted in AWS_DELTA_LAKE.md section 10

Delta or Iceberg

03



Where Delta and Iceberg plug in, and where they don't

Console and UI · terminals, pages, BI reports, Admin

Server · BI, VaR and exposure engines, captures, Drishti*net (replicas, shards, forwarding)

Packs and data contracts · kinds, promoted columns, JSON Lines in · connector names (trade-calypso)

Engine · SourceRouter: a kind to its connectors, in order; reads, searches, columns, history, scans

Identical on both formats

Nothing above the line knows whether a table is Delta or Iceberg. A pack routes kind `trade` to `trade-calypso`; the file under that name decides the format.

The seam · drishti-api `SourcePlugin`: fetch, days, columnar, columns, scan, dataVersion, heldDates, vectors, documents, tableIdentity …

plugin: delta

- Delta Kernel over `drishti-deltalake` (native) or Hadoop
- tables Drishti or others write
- vectors, whole-day documents

plugin: iceberg

- `drishti-iceberg` (native, default) or Apache Iceberg Java (library)
- Drishti's layout: id, doc, promoted
- loader and maintenance write

plugin: iceberg-external

- `drishti-iceberg` only
- anyone's tables, the row is the document
- read-only

other plugins

- files, S3 objects, PostgreSQL, DuckDB, Cassandra, Aerospike, MongoDB, Redis …

drishti-lakeio (shared): local disk, S3, byte ranges, codecs, footer cache, exclusive create, conditional PUT

Format-specific below the seam only

Storage · local disk or EFS (development, Windows) · Amazon S3 (`s3://`, `s3a://`) · Azure, GCS, HDFS only through the Hadoop/library engines

What is identical, what differs, and why

Layer	Identical on both	Differs	Why it differs
Packs, contracts, connector names	kinds, promoted columns (`layout.<kind>.columns`), JSON Lines input, routes	nothing	the format is below the plugin
Logical table layout	`id`, `doc`, `business_date`, promoted columns; sorted by id; one partition per day; 250,000 rows a file	row groups by rows (Delta, 1,000 rows or 1 MB on AWS) vs by compressed bytes (Iceberg, 1 MB)	Iceberg's writer sizes row groups in bytes
Read paths	id map, one row group per read, column sets, type-ahead, history, BI column scan	Iceberg caches by day fingerprint; Delta by table version	Iceberg's manifests give each day's files cheaply
Where the table's truth is	a folder in S3, Parquet data files	`_delta_log/` commits and checkpoints vs `metadata/` JSON, manifest lists, manifests	the two specifications
Commits on S3	a conditional PUT (`If-None-Match: *`): two writers never both win	Delta: `lake publish` (Python deltalake); Iceberg: the Java loader itself, plus a probe	Drishti's Java Delta writer writes local disk and EFS only
Catalog	none needed (path-based)	Iceberg may also use a REST catalog (Glue, Polaris)	Iceberg's ecosystem is catalog-centred
Maintenance	`lake maintain` and Admin → Lakes for both; retention, relayout, 7-year floor	Delta checkpoints and vacuum vs Iceberg snapshot expiry and orphan files	the two specifications keep history differently

Drishti's support, side by side

Capability	Delta Lake (default)	Iceberg
Documents (trades, reference, public data); local disk, EFS, S3 without Hadoop; Windows	yes	yes
Write a table in S3	yes (`lake publish`)	yes (the loader, directly)
Tables other engines write, in their own layout	yes (Delta connector)	yes (`iceberg-external`, read-only)
VaR inputs, intraday files, Kafka windows; exposure inputs (`exposure-path`)	yes	yes (equal to the cent)
VaR sharded across servers; the parallel Java loader (folder or S3)	yes	yes
Official results, folder or `s3://`; BI readers of them	yes; BI follows the format by itself	yes; BI follows the format by itself
Load and maintain	`lake build`, `publish`, `maintain`	`lake build --format iceberg`, `maintain`
Maintenance UI, schedules, ownership rule, 7-year risk floor	yes	yes (+ orphan files)
Disk caches (id maps, documents) across restarts	yes	yes
Caches survive a commit of another day	PROPOSED	yes
AWS scripts: load, backfill, maintain, lifecycle, alarms, report (dry-run tested)	`deploy/aws/delta-lake/`	`deploy/aws/iceberg/`
Catalog	the table's log in S3 (none needed)	path-based in S3; REST, Glue, S3 Tables client (on AWS: NOT RUN)
A promoted column added	new table + backfill; in place: PLANNED	in place, by field id
Conversion between the two (any table)	PLANNED	PLANNED

The seam in detail: what each plugin implements today

Capability (SourcePlugin method)	delta	iceberg	iceberg-external	Who needs it
one entity, history days, search, reverse links (`fetch`, `days`, `search`, `reverse`)	yes	yes	yes (no `days`)	every page
columns of a day (`columnar`, `columns`), type-ahead index	yes	yes	yes (columns)	searches, pick lists, impact
BI column scan, data version, held dates, row estimate	yes	yes	—	BI store and runs, statements
time travel (*known at*)	yes	yes	yes	every as-known-at view
vector columns (`vectorColumns`, `vectors`): `list<double>` streamed	yes	yes	—	VaR, exposure
whole-day documents (`documents`)	yes	yes	—	derived pages (faster path)
table identity, holds none (`tableIdentity`, `holdsNone`)	yes	yes	—	Drishti*net replicas vs shards
inventory, refresh on signal (`inventory`, `refreshVersions`)	yes	yes	—	Admin test, data-load signal

Clean layering, equal capability. The gaps were implementations of existing interface methods, not changes to the architecture: the seam was already where it should be.

Switching one table, and running both side by side

```
python tools/build_demo_lakes.py --store calypso=iceberg      # CALYPSO in Iceberg, ENDUR and MUREX in Delta
python tools/build_demo_lakes.py --store all=iceberg          # every system in Iceberg
```

Before: every system in Delta

trade-endur · `plugin: delta` · `…/trade_endur`

trade-calypso · `plugin: delta` · `…/trade_calypso`

trade-murex · `plugin: delta` · `…/trade_murex`

After: CALYPSO in Iceberg, the same connector name

trade-endur · `plugin: delta` · `delta/trading/trade_endur`

trade-calypso · `plugin: iceberg` · `iceberg/…/trade_calypso`, native

trade-murex · `plugin: delta` · `delta/trading/trade_murex`

Compared, all-Delta against mixed (`stores_equality.py`)	Result
six trade pages; `TRD <id> 10` histories for each	the same
two counterparty-trades pages; three searches	the same
a BI model query	the same
total	**18 answers the same**

What stays: the trading pack's route `trade → trade-endur, trade-calypso, trade-murex`, the kinds, the promoted columns, every page, search, history and BI model; BI column scan, data versions and held dates on both.

What changes: one connector file's `plugin`, `engine` and `table`.

Never two connectors for one system's trades: the union would double them.

Inside each format: a log against a tree

Delta Lake: a transaction log

table folder

- `s3://…/lake/trading/trade_endur/`

_delta_log/

- `…0451.json` `…0452.json` one commit each: add / remove + stats
- `…0450.checkpoint.parquet` every 10 commits
- `_last_checkpoint`: where to start replaying

data files

- `business_date=…/PRODUCT_TYPE=…/part-….snappy.parquet`

Read: replay checkpoint + later commits → the file list. Commit: create the next `NNN.json`.

Iceberg: a metadata tree

catalog pointer

- `version-hint.text` (path) or the REST catalog's metadata location

vN.metadata.json

- schemas, partition specs, sort orders, snapshots, properties

manifest list (per snapshot)

- `snap-….avro`: the manifests, with partition summaries

manifests

- `…-m0.avro`: data and delete files, partition values, column bounds

data and delete files

- `data/business_date=…/…parquet`; position or equality delete files

Read: pointer → metadata → one manifest list → pruned manifests. Commit: create `vN+1.metadata.json` (or the catalog swaps the pointer).

Side by side (1/3): log, commits, time travel, evolution

Aspect	Delta Lake	Apache Iceberg	In Drishti today
Where the truth is	an ordered log of JSON commits + Parquet checkpoints	a tree: metadata JSON → manifest list → manifests	both read natively, no Hadoop
Commit on S3	create `_delta_log/N.json` if absent	create `vN+1.metadata.json` if absent, or a catalog swap	both by `If-None-Match: *`; Delta via `lake publish`, Iceberg via the Java loader
Concurrency	optimistic; conflicting writers retry	optimistic; requirements re-validated	a rewrite of a changed day is refused (Iceberg); per-table locks in the scripts (Delta)
Time travel	by version or timestamp, until vacuum and log cleanup	by snapshot or timestamp, until snapshots expire	*known at* on both; 7-year floor for risk tables on both
Schema evolution	add columns by name; renames need column mapping	by field id: add, drop, rename, reorder, widen	Iceberg reads every change by id; Delta publish refuses a changed day (in-place add: planned, not built)
Reading cost of a refresh	commits since the last checkpoint	one small pointer, then metadata if changed	Iceberg keys caches by day

Side by side (2/3): partitions, deletes, maintenance, catalogs

Aspect	Delta Lake	Apache Iceberg	In Drishti today
Partitioning	Hive-style folders on columns; liquid clustering (Databricks)	**hidden**: transforms (day, bucket, truncate) in the spec; specs evolve	both: `business_date` only (+ PRODUCT_TYPE on Delta/AWS); Iceberg transforms read
Row deletes, merges	deletion vectors or file rewrites (copy-on-write)	position / equality delete files (merge-on-read) or rewrites	Drishti writes neither: a day is replaced; reads others' deletes (Delta: columns off; Iceberg: columns on)
Maintenance	OPTIMIZE, checkpoint, VACUUM	rewrite data and manifests, expire snapshots, remove orphans	both: `lake maintain` + Admin → Lakes; Iceberg with orphan removal
Catalog	optional (the log is the table); Unity, Glue for discovery	central to most engines (Glue, Polaris, Unity, Nessie); path tables work	Delta: none; Iceberg: path or REST, incl. Glue and S3 Tables (not yet run on AWS)
Table identity across nodes	table id in the log	table UUID in metadata	both report it to Drishti*net

Side by side (3/3): the ecosystem and the running cost

Engine	Delta Lake	Apache Iceberg
Apache Spark, Amazon EMR	read and write (Delta's Spark library)	read and write (Iceberg's Spark runtime)
Trino, Starburst	Delta connector	Iceberg connector
Amazon Athena	reads Delta tables	reads and writes Iceberg tables in the Glue catalog; OPTIMIZE, VACUUM
Snowflake	reads through external tables / catalog integration	native Iceberg tables (managed or catalog-linked), Open Catalog (Polaris)
Databricks	native (Unity Catalog)	through Unity's Iceberg REST interface or UniForm
pyiceberg, DuckDB, pandas	delta-rs (deltalake), DuckDB's delta extension	pyiceberg, DuckDB's iceberg extension (both used in Drishti's tests)

Running cost item	Delta	Iceberg
Storage (Parquet), tiers	same	same
Metadata per commit	a small JSON; a checkpoint every 10	a metadata JSON listing every snapshot kept + a manifest list + a manifest
Catalog	none	none (path) or Glue / S3 Tables request and storage charges
Maintenance compute	nightly `lake maintain`	nightly `IcebergMaintenance`, or managed optimizers (charged)

drishti-iceberg: Drishti's own Iceberg engine, no Hadoop

IcebergSourcePlugin · IcebergExternalPlugin · IcebergLoader · IcebergMaintenance	unchanged read paths
catalog/ path-based (`version-hint.text`), a metadata file named directly, REST (OAuth2, SigV4, vended credentials)	Jackson, JDK HTTP
meta/ table metadata v1/v2, schemas, partition specs, sort orders, snapshots, refs; unknown fields kept	Jackson
manifest/ manifest lists and manifests: data and delete manifests, partition summaries, column bounds	Avro
scan/ + read/ pruning by partition and bounds; Parquet by field id; position and equality deletes	parquet-java
write/ Parquet with field ids and metrics, manifests, commits, maintenance operations	no Hadoop Configuration
drishti-lakeio local disk (Windows paths, UNC), S3 (AWS SDK v2), ranged reads, codecs, footer cache	shared with Delta

0 Hadoop classes

loaded by connector, loader or maintenance (a test refuses them)

Windows

no winutils.exe, no HADOOP_HOME: load-iceberg.ps1, start-server.ps1 -Iceberg

The default engine

native; the library kept for Azure, GCS, HDFS

Refused by name, never read wrongly: format version 3, encrypted tables, ORC and Avro data files, unknown types, S3 remote signing.

Two engines: native and library

engine	What reads and writes	Reaches	Use it for
`native` **(default)**	drishti-iceberg, no Hadoop	local disk (Windows too), `s3://`, `s3a://`; path, named metadata, REST	everything on disk and S3; Windows
`library`	Apache Iceberg Java 1.10.1 over Hadoop	the above plus `abfs://`, `gs://`, `hdfs://`	Azure, GCS, HDFS; a fallback
`auto`	native on Windows, library elsewhere		keeping the library on Linux

Measured (10,000 trades × 3 days)	native	library
load (30,000 rows)	22.7 s	25.7 s
connector start (plan, type-ahead)	637 ms	1,848 ms
first trade (cold: id map, footer)	212 ms	288 ms
one trade, document cache emptied: median / p95	54 / 111 ms	75 / 144 ms
a trade on a past date (day not yet read)	94 ms	353 ms

Why native is faster on first reads: each Parquet footer, manifest and delete file is read once and kept (data files never change). The library re-reads the footer on every single-entity read.
Caches: footers 50,000 row groups (shared with Delta), manifests 2,000,000 entries, delete rows 5,000,000.

Two connectors: Drishti's tables and anyone's tables

	`iceberg` (Drishti's tables)	`iceberg-external` (anyone's tables)
Written by	`IcebergLoader` in Drishti's layout	Spark, Flink, Trino, Snowflake, Glue, dbt, pyiceberg, any layout
The document	the `doc` column, exactly as loaded	**the row** as JSON (structs, lists, maps), or a JSON column
The id	the `id` column	one or several columns; by default the table's **identifier fields**
The business date	`business_date`	a date column, a timestamp's day in a zone, the partition spec's date; or current only
Searchable fields	the promoted columns the pack declares	**every primitive column** (nested to depth 2)
One entity	id map: one row group	sorted or bucketed by id: one row group; unsorted: an **id index** built on first use per day
Writes	loader and maintenance	**never**: the owners' engines maintain the tables
Deletes, time travel	position and equality deletes applied; *known at*	the same; branches and tags not read

Catalogs: what Drishti supports today

Catalog	How Drishti reaches it	Writes	Status
Path-based (`catalog: hadoop`, the default)	`<root>/<domain>/<table>/metadata/version-hint.text`; root a folder or `s3://`	loader and maintenance	
A metadata file named directly	`metadata-location.<kind>: s3://…/v12.metadata.json`	read-only	
REST: Polaris, Snowflake Open Catalog, Nessie, Lakekeeper, Unity, Gravitino	`catalog: rest`, `uri`, `warehouse`, OAuth2 `credential` or `token`	commits with requirements	
AWS Glue Data Catalog, through Glue's Iceberg REST endpoint	`uri: https://glue.<region>.amazonaws.com/iceberg`, SigV4 `signing-name: glue`; `check-glue.sh`	as REST	**NOT RUN** on AWS
Amazon S3 Tables (table buckets)	its Iceberg REST endpoint, SigV4 `signing-name: s3tables`; `check-glue.sh --s3-tables ARN`	as REST	**NOT RUN** on AWS
The Glue and S3 Tables live runs	`GlueLiveTest` with `DRISHTI_GLUE_LIVE=1`, in the product owner's account	—	**PLANNED**
AWS Glue Data Catalog API (GetTable, UpdateTable); Hive metastore, JDBC	not needed: Glue's REST endpoint; others behind a REST catalog	—	via REST only

```
settings: { catalog: rest, uri: "https://glue.us-east-1.amazonaws.com/iceberg", warehouse: "<account-id>",
            catalog.rest.sigv4-enabled: "true", catalog.rest.signing-name: glue, catalog.rest.signing-region: us-east-1 }
```

The catalog choice on AWS

Path-based tables in S3

- no catalog service: `metadata/version-hint.text` in the table folder
- commits: conditional PUT, probe at first commit
- other engines read by naming the metadata file, or after registering the table in Glue
- **best when** only Drishti writes and Drishti reads

AWS Glue Data Catalog (Iceberg REST endpoint)

NOT RUN

- `catalog: rest`, SigV4 `signing-name: glue`; `ICEBERG_CATALOG=glue` in the scripts (built; not yet run on AWS)
- commits through the catalog (409 retried); Athena, EMR, Redshift see the same tables
- IAM (and Lake Formation, if used) governs every table
- **best when** Athena or Spark also read or write

Amazon S3 Tables (table buckets)

NOT RUN

- its Iceberg REST endpoint, SigV4 `signing-name: s3tables`, a table bucket ARN (written; not yet run on AWS)
- `check-glue.sh --s3-tables ARN`: the live run is the product owner's, not yet done
- AWS runs compaction and snapshot management itself
- **prove first** in a development account

Recommendation (PROPOSED): start path-based (built, no extra service, the same commit safety as Delta). Move a table behind Glue's REST endpoint when another engine must write it or Athena must query it; the connector's name, kinds and packs do not change, only `catalog`, `uri` and the signing settings. Run `check-glue.sh` first.

Schema and partition evolution, as built

Read by field id

renamed columns read correctly; added columns read as null; int→long, float→double, decimal widening; files without ids through `schema.name-mapping.default`

Add a promoted column

the loader adds new promoted paths as columns; `IcebergMaintenance --columns trade:PATH` adds and fills them from the documents

Partition evolution (read)

several specs in one table; every transform: identity, bucket, truncate, year, month, day, hour, void

Hidden partitioning (read)

a table partitioned by `day(TRADE_TS)` or `bucket(id)` is pruned by its manifests; `iceberg-external` takes the business date from the spec

Drishti's writer changes a spec

always `identity(business_date)`; no partition evolution written, none needed

Versus Delta in Drishti: adding a promoted column on Delta changes the table's columns, and `lake publish` then refuses the next day: a new folder (`trade_endur_v2`) and a backfill (Delta's in-place add: planned, not built).
On Iceberg the column is added in place by field id; old files read it as null until `--columns` rewrites them.

Why it matters for a bank: promoted columns grow as people search new fields. Iceberg absorbs that without a new table; Delta in Drishti's AWS flow needs a new table per change, because `lake publish` refuses a day whose columns differ.

Snapshots, time travel and deletes

A table's snapshot log, and what *known at* reads



A view *known at* that instant reads S3's schema and files; a later load is invisible; a time before the oldest snapshot kept reads nothing.

	Iceberg in Drishti	Delta in Drishti
Time travel unit	a snapshot (each commit), from the snapshot log	a table version (each commit), from the log
How far back	until snapshots are expired: `--expire-hours` (168 by default)	until vacuum and log retention (168 h; 7 years in the risk domains)
Rows deleted by other engines	position and equality deletes applied by **every** read, columns stay on	deletion vectors honoured; the table's searches fall back to documents
Drishti's own deletes	none: a restated day replaces its partition	none: a restated day replaces its partition
Format v3 deletion vectors	refused by name (later)	—

Lake build to Iceberg, and the demo's CALYPSO in Iceberg

1 JSON Lines in

`data/testdata/<domain>/<kind>/*.jsonl.gz` or a bank's dumps

2 lake build → Delta

staged under `<data>/build/iceberg-stage`: the same documents and promoted columns as a Delta build

3 IcebergLoader, engine native

each business day committed as one overwrite of its partition; no Maven, works on Windows

4 Iceberg table

`<data>/iceberg/trading/trade_calypso` (or `s3://…`)

5 connector file switched

`trade-calypso.yaml`: `plugin: iceberg`, `engine: native`, `table: {trade: trade_calypso}`

```
tools/load-iceberg.sh ./data/iceberg --trades 10000 --days 3          # samples and a book, one load
tools/load-iceberg.sh s3a://risk-lake/iceberg --set s3.region=us-east-1 --spill-dir /data/tmp
python tools/lake/stores.py --from data/delta/trading/trade_calypso --root data/iceberg --table trade_calypso --replace
```

Path to an Iceberg table	Scope
`build_demo_lakes.py --store <system>=iceberg` (via `stores.py`)	the trading domain's trade tables
`load-iceberg.sh` / **`load-iceberg.ps1`** → **`IcebergLoader`**	any kind, JSON Lines `{domain, kind, id, date, doc, columns}`
`lake build --format iceberg`; **`--store risk=iceberg`** (the whole risk domain)	any pack and kind; vector columns kept; `s3://` or a REST catalog

A decision guide for a bank

If the bank ...	Choose	Because (Drishti today)
has no format standard yet, or runs Kafka intraday heavily	**Delta** (the default)	the default everywhere; small JSON commits; the longest-run AWS scripts and runbooks
runs VaR or exposure and standardises on Iceberg	**Iceberg** for the risk domain	`--store risk=iceberg`: VaR, exposure, intraday and official results equal Delta's to the cent
already keeps trades in Iceberg (Snowflake, Glue, Spark pipelines)	**Iceberg**, read in place	`iceberg-external` serves them read-only; no copy, no Drishti writer
needs other engines to write the same tables Drishti reads	**Iceberg** behind a REST catalog	commits through the catalog's requirements; Drishti's loader is one writer among several
changes promoted columns often	**Iceberg** for those tables	a column is added by field id; Delta's publish needs a new table
standardises on Databricks and Unity Catalog	**Delta**	Drishti reads Delta natively; Iceberg only through a REST catalog
must run Drishti on Windows laptops for QA	either	both native engines run without Hadoop

The recommendation, in one page

Delta by default; Iceberg wherever the bank standardises on it

- **Risk domain** (P&L vectors, scenario PVs, exposure paths, official results): Delta by default; Iceberg equal to the cent
- **Trades, reference, market data**: either, chosen per source system (`--store`)
- **The bank's Iceberg tables** (Snowflake, Glue): `iceberg-external`, read in place
- **One switch per table**: the connector name never changes, so a table can move later

Built for Iceberg; what remains

1. `vectors()`, documents, identity, disk caches; VaR, intraday, exposure on Iceberg
2. Official results to Iceberg and to `s3://` (both formats); BI readers follow
3. `deploy/aws/iceberg/`, orphan files, Admin → Lakes, the 7-year floor
4. VaR sharded across a cluster and the parallel Java loader, on either format
5. **NOT RUN:** the Glue and S3 Tables runs on a real AWS account (`check-glue.sh`)
6. **PLANNED:** Delta ↔ Iceberg conversion, Delta in-place schema evolution, the 200K benchmark

Converting tables between formats

Direction	How	History	Status
Delta → Iceberg (trade tables)	`stores.py` / `build_demo_lakes.py --store <system>=iceberg`: rows → `IcebergLoader`	current days; one snapshot per copied day	
Risk domain, either way	`build_demo_lakes.py --store risk=iceberg` (or `risk=delta`): rebuilt from JSON Lines	the days in the JSON Lines	
Any table, Delta ↔ Iceberg	a conversion of an existing table in either direction	current days	**PLANNED**
In place: Delta UniForm, Iceberg `snapshot`/`migrate`	other engines write the second format's metadata over the same Parquet	the converter's choice	**PROPOSED** (evaluate)

Keep the old table until the new one is proven: build the Iceberg table beside the Delta one, run the equality test against both, then switch the connector file. Keep the Delta folder for its time travel (and the content store keeps every official capture), or archive it.
Never let two connectors serve one system's trades at once: the union would double them.

Keeping one logical connector name across a switch

1 Build beside

write the Iceberg table next to the Delta one (`stores.py`, or the loader)

2 Compare

`stores_equality.py`-style answers, both lakes, the same dates

3 Switch

edit `trade-calypso.yaml`: `plugin: iceberg`, `engine: native`, `root`, `table.trade`, the same `layout`

4 Signal

the data-load signal, or wait `refresh-seconds`; Health shows `engine: native`, UP

5 Keep, then retire

keep the Delta table for its history; roll back by restoring the file

```
# config/connectors/trade-calypso.yaml (the name never changes)
plugin: iceberg                 # was: delta
kinds: [trade]
settings:
  domain: trading
  engine: native
  root: ${DRISHTI_ICEBERG_ROOT:${drishti.data.dir:./data}/iceberg}
  table: {trade: trade_calypso}
  layout: {trade: {columns: [TRADE_ID, SOURCE_SYSTEM, …]}}
```

Nothing above the connector moves: the pack's route, saved searches, BI models, reports and bookmarks name the connector or the kind, never the format.
Caches of the old connector are dropped when the file changes. At an upgrade the edited file keeps its values and gains the pack's new keys.

The risk lake

04

The risk lake's tables: inputs in risk/, results in risk-results/

Table	One row is	Key	Vector column	Read by
risk/pnl-vector	a trade's (or position's) P&L over 509 scenarios	PNL-<trade>-<set>	PNL	VaR engine
risk/scenario-pv	a trade's scenario PVs and base PV, its own currency	SCENARIO_PV_ID	SCENARIO_PV	VaR engine
risk/fx-scenario	a currency's rate today and per scenario	FX_SCENARIO_ID	none (RATES in doc)	VaR engine
risk/scenario-set · **risk-hierarchy** · **portfolio**	the scenario days · a tree node with its limit · a portfolio	SCENARIO_SET_ID …	none	VaR engine
risk/var-limit	a node's VaR limit in a hierarchy by columns (the limit letters)	VL-<NODE_ID>	none	VaR engine
risk/simulation-run	an exposure run's grid and identity	RUN_ID	none	exposure engine
risk/exposure-path	a trade × grid date: MtM over the P paths	PATH_ROW_ID	MTM	exposure engine
risk-results/var-official	a node × date × result version	node@date#v	none	BI only (VOFF)
risk-results/exposure-official(-cpty)	a node, set, counterparty or group × version	scope@date#v	none (profile as JSON)	BI only (XOFC)

Engines read risk/ only; nothing reads risk-results/ back except BI's read-only connectors (a guard test). Every input row carries LIMIT_LETTER. Positions are derived, never stored.

A vector is one list<double> column: why, and how it is carried

Layout (one date, 2M rows × 509)	Size, compressed (estimate)	Sum a scope's vectors	Verdict
A. Long: a row per position × scenario	~9–11 GB[1]	a group-by over hundreds of millions of rows	rejected
B. Wide: 509 columns	~5.5–6.5 GB	509 column chunks per row group	rejected (schema = scenario count)
C. Vector: `PNL list<double>`	~5.5–6.5 GB	one column, decoded into one array, added	**chosen, built**
D. Blob: 4,072 bytes	~6–7 GB	one column, copy bytes	rejected (other tools cannot read it)

As built

- `layout.<kind>.vectors: [PNL]`; `lake build --vector`; both writers
- the document keeps `{"$vector": "PNL"}`: never stored twice
- streamed per date, one buffer reused for every row
- round trip JSONL → Delta → `double[]` equal to the bit
- `list<float>` read and written for exposure paths

Where it runs

- streamed reads: the Delta plugin (default) and the Iceberg plugin (`SourcePlugin.vectors`)
- VaR and exposure on Iceberg: equal to Delta to the cent
- Iceberg loader: `list<double>`; a missing element null, never 0; a non-number list a reject
- bank-scale layout (sorted by legal entity and desk, BSS + ZSTD): the parallel Java loader writes it

[1] design estimates of VAR_GUIDE.md 29.1 for 2,000,000 rows × 509 × 8 B = 8.14 GB raw: vector and wide ≈ 0.68–0.8 × raw compressed; long adds an id and scenario index per value; ESTIMATED, not measured

Morning run, intraday batches, Kafka micro-batches: all into the lake

Morning run · RUN_KIND FULL

- every live trade, one BATCH_ID
- JSONL → `lake build` replaces the date

Intraday files · INTRADAY

- changed trades only, own BATCH_ID
- `drishti.py var intraday --load` (append)

Kafka PV topic · micro-batches

- `pv-updates` (var-pv-stream), JSONL text
- window 5 min · flush at 500 · Apply now

risk/ in Delta or Iceberg (date D)

- the morning commit replaces D
- each intraday batch: one appended commit
- each Kafka window: ONE batch, `K-<date>-<time>-<n>`
- latest `COMPUTED_AT` per trade wins
- the lake is the record: a restart rebuilds
- 7 years of time travel

signal

VaR engine (one date hot)

- full run: date reloads
- batch: new − old into the book and every node above
- official capture from the morning run only

risk-results/var-official

- append-only, a row per node and version
- JSONL beside it
- a folder or `s3://`
- read by BI as `VOFF`

Measured: 200 batches of 3 trades[1] equal one full load to the cent, on every node; batch time does not grow.

[1] measured: VarIntradayApiTest (VAR_GUIDE.md 8.4)

On AWS: who writes the risk lake, and where

Table group	Written by	On an S3 lake today
`risk/` morning runs (files)	the bank's batch: JSONL → `lake build` → `lake publish` (overwrite-dates)	load-day.sh, like any table
`risk/` intraday files	`lake build --mode append` → `lake publish --mode append`	the tools; no script yet
`risk/` Kafka windows	the server's Java lake writer (`pv-updates' lake`)	local or EFS lake path
`risk-results/` official	the server's publisher, Delta (default) or Iceberg, + JSONL	written to `s3://` itself

Risk tables in S3 keep 7 years: publish them with `--log-retention-days 2557 --deleted-file-retention-days 2557`; `lake publish`'s defaults (30 and 7 days) would cut time travel.

Official results to S3

- `drishti.var.official.root: s3://<lake>/risk` (and the exposure root); `format: delta` or `iceberg`
- each Delta commit an exclusive create: two servers both land
- BI's `var-results` follows the format; task role: `s3:PutObject`, no delete

Kafka windows to S3

PROPOSED

- `pv-updates`' lake on EFS; each window published with `--mode append`
- or keep `risk/` on EFS if intraday must be seconds

Hot files: `DRISHTI_VAR_HOT_DIR`, `DRISHTI_EXPOSURE_HOT_DIR` on fast local disk, as the guides say (slide 22).

Official results: append-only, versioned, reproducible

Column group	Content
`business_date` · `id` · `doc`	partition; `<NODE_ID>@<date>#v<RESULT_VERSION>`; the row as JSON
node	`HIERARCHY` (MGMT, LEGAL, LL …), `NODE_ID`, `NODE_NAME`, `PARENT_ID`, `LEVEL`
version and inputs	`RESULT_VERSION` (1, 2 … on restatement), `CAPTURE_ID`, `INPUT_VERSIONS` (versions, batch ids, rules digest)
figures	`VAR`, `ES`, `CONFIDENCE`, `ES_CONFIDENCE`, `STANDALONE_SUM`, `DIVERSIFICATION`, `HYPO_PNL`
limits and backtest	`LIMIT`, `LIMIT_USE`, `EXCEPTIONS_250`, `TRAFFIC_LIGHT`, `TRADE_COUNT`, `POSITION_COUNT`

- every configured hierarchy: 131 nodes on the test data (MGMT, LEGAL + 43 limit-letter nodes)
- `delta.appendOnly`: nothing is overwritten; a restated date adds rows with the next `RESULT_VERSION`
- 7 years of log and removed files: time travel to any published day
- no vectors, no trade rows: the capture in the content store keeps the vector for audit
- Delta or Iceberg, a folder or `s3://`

For other systems, never for engines. BI reads the published tables as `VOFF` and `XOFC` (the reports Market risk: VaR and Counterparty risk; no measure sums VaR). Each figure can be recomputed from its inputs, the same vectors and engine version, years later.

Official results: Delta or Iceberg, a folder or S3, JSON Lines beside


```
<root>/risk-results/                          a folder or s3://
├── var-official/              business_date=…/   Delta (default) or Iceberg, append-only, 7 years
├── exposure-official/  exposure-official-cpty/   append-only
└── jsonl/<table>/<date>/<table>-v<table version>.jsonl      one file per commit
```

Setting	Default	Effect
drishti.var.official.root	${DRISHTI_DELTA_ROOT}	where both are written: a folder or s3://
drishti.var.official.format	delta	or iceberg; a table keeps the format it was created in
drishti.ccr.exposure.official.format	delta	the same for exposure
….official.storage	—	the s3.* settings (no keys: the task role)
drishti.var.official.jsonl	true	each VaR publish also written as JSONL beside the table
drishti.ccr.exposure.official.jsonl	true	each exposure publish also written as JSONL

- **Append-only on both:** delta.appendOnly; Iceberg append commits only (drishti.append-only=true), a snapshot per run
- **7 years on both:** Delta log retention; Iceberg history.expire.max-snapshot-age-ms, every metadata file kept
- **On S3:** each Delta commit an exclusive create: two servers publishing at once both land
- **BI's readers** (VOFF, XOFC) switch to plugin: iceberg by themselves

- A restatement is a new file, never a rewrite; content-store row lists are JSONL too (var/official/<date>-membership-v<n>.jsonl); older .json records are read as they are
- Tested: OfficialSinkTest, OfficialS3Test (S3Mock), OfficialReadersTest; not yet run against a real bucket

Exposure paths: the large table, and what is kept for 7 years

Item (1,000,000 trades)	P = 2,000, G = 50	P = 5,000, G = 100
trade paths, float32, dense, a date	400 GB[1]	2.0 TB
the same, float64	800 GB	4.0 TB
netting-set V matrices (hot), 100,000 sets	40 GB	200 GB
for comparison: VaR vectors, a date	4.07 GB	

What	Kept (decided 2026-10-26, built)
trade paths	20 business days, and each month-end for 1 year
netting-set matrices	every date for 1 year; month-ends for 7 years
statistics per set and counterparty	every date, 7 years
trade EE contributions	every date, 7 years
official captures, run records	7 years

3.5 PB → about 130 TB[2]: 7 years of dense trade paths at P = 5,000, G = 100 would be 3.5 PB; under the decided tiers the standing storage is about 130 TB at full size (a fifth at P = 2,000, G = 50). ESTIMATED from the guide's formulas.

Exposure-path layout at scale (EXPOSURE_GUIDE.md 15.3) PROPOSED

- partition by `business_date` and `run_id`; sort by `NETTING_SET`, `TRADE_ID`, `GRID_INDEX`; ~512 MB files; `MTM` BYTE_STREAM_SPLIT + ZSTD

[1] trades × P × G × 4 B (EXPOSURE_GUIDE.md 27.2) [2] 2.0 TB × 1,764 dates; kept tiers summed = 132 TB Full derivations: slides 133–136.

Evidence

05

Status: what exists today, and how it was tested

No label: built, in the code and tested: the AWS lake, the risk lake, JSON Lines, the S3 work, official results to S3, Iceberg as an alternative, Drishti*net, VaR sharding and the parallel Java loader.

DESIGNED — The super-scale scale-out of the whole lake (SUPER_SCALE S1–S8), not built.

PLANNED — In the plan, not built: Delta in-place schema evolution, Delta ↔ Iceberg conversion, a bank-scale benchmark on both formats, Unity Catalog.

NOT RUN — Code written, never run against AWS: the Glue and S3 Tables live runs.

PROPOSED — This deck's recommendation, not yet in the docs (each one is marked where it appears).

MEASURED — A number measured on Drishti (the doc names the test or benchmark).

ESTIMATED — A number from arithmetic or a linear extrapolation of measured points; to be replaced by measurement.

How it was tested

Every AWS script, Delta and Iceberg, with --help and --dry-run (`tools/test_deploy_aws.py`, an aws that fails if called). Delta on S3 and `lake publish` on moto, three publishes at once; official results to S3 on S3Mock in both formats (`OfficialS3Test`); VaR and exposure equal on Delta and Iceberg to the cent (`VarFormatEqualityTest`, `ExposureFormatEqualityTest`); sharded VaR equal to one server (`VarShardClusterTest`); the connector upgrade (`ConnectorUpgradeTest`).

Not yet: the first run in a real AWS account (steps 0–11 and the runbooks); the Glue and S3 Tables live runs. Every number marked ESTIMATED waits for it.

Built (1/2): the lake on S3

Capability	Where
Reading Delta tables in S3: file lists from the last checkpoint, id maps, one row group per read, footers cached, documents cached on disk	drishti-deltalake, drishti-lakeio, plugins/drishti-plugin-delta
`lake build`: JSON Lines into a Delta table on local disk or EFS: promoted and vector columns, partitions, sorted by id per batch	tools/lake/build.py, the Java writer in the server jar
`lake publish`: a built table committed into an S3 table as ONE commit (S3 conditional writes)	tools/lake/publish.py, test_publish.py
`lake maintain` on `s3://`: retention, checkpoint, vacuum; relayout of the per-source trade tables	tools/lake/maintain.py (`--force-foreign`)
Buckets, IAM, conditional-write check, lifecycle tiers, alarms, report, load, backfill, maintenance, EFS, Cloud Map, task definitions, schedules	deploy/aws/ (dry-run tested)
ECS images (server, console, loader) and entry points	deploy/docker/ecs/ (files)
Drishti*net shards by table; replicas send one entity to the same node by a hash of its id	DRISHTI_NET.md §7, §10

Built (2/2): history, risk, Iceberg, sharding, the loader

Capability	Where
History queries over the lake (`TRD <id> 30`, `knownAt`), the Delta fast path, across Drishti*net	HISTORY_QUERIES.md, DELTA_CONNECTOR.md §8
Delta and Iceberg side by side, per source system; Drishti*net gossip on its own port, peer data over HTTP/2	build_demo_lakes.py `--store`; NET.md §4, §7
Content store with an S3 backend (conditional PUT, ETag); S3 on S3Mock, moto, MinIO	CONTENT_STORE.md, TOOLS.md §4.12
Vector columns `list<double>`; VaR and `exposure-path` inputs in `risk/`; 7-year floor; official results append-only	VAR_GUIDE.md §15, §19, §29; EXPOSURE_GUIDE.md §15
JSON Lines standard; CSV deprecated; official results as JSONL; scenario PVs; intraday and Kafka batches; DATAMAP	DATA_FORMATS.md, VAR_GUIDE.md §8.4
Official results straight to `s3://`, Delta (default) or Iceberg; each Delta commit an exclusive create	VAR_GUIDE.md §19, AWS_DEPLOYMENT.md §22
Iceberg a full alternative: VaR, intraday, exposure, official; `deploy/aws/iceberg/`	ICEBERG_CONNECTOR.md §18–19
Connector upgrades: unedited files follow the pack, edited ones gain keys; hierarchies by columns, `var-limit`	CONNECTOR_FILES.md, VAR_GUIDE.md §5.2
VaR sharded across Drishti*net: shards of legal entity and desk ranges, two owners each, partial vectors added	VAR_SHARDING.md
The parallel Java loader: JSON Lines (folder or S3) into Delta or Iceberg, one commit per date, shard-aligned files	PARALLEL_LOADER.md

Iceberg as built, and where each piece is tested

Piece	Where, and how it is tested
drishti-iceberg, no Hadoop: metadata, manifests, planning with pruning, field-id reads, deletes, time travel; the writer (Parquet with field ids and metrics, manifests, commits)	`drishti-iceberg`, `drishti-lakeio`; pyiceberg interop both ways; a class loader that refuses Hadoop; `NativeCommitTest`
`iceberg` connector: id maps, column sets, type-ahead, history days, BI scan, data versions, *known at*; `vectors()`, `documents()`, `tableIdentity()`, `inventory()`, `refreshVersions()`; disk caches	`plugins/drishti-plugin-iceberg`; every plugin test on both engines; `IcebergParityTest`, `IcebergVectorsTest`, `IcebergDiskCacheTest`
`iceberg-external`: anyone's tables, read-only; rows as documents, identifier fields, every column searchable, id index, `tables: auto`	`IcebergExternalPluginTest` on pyiceberg-written fixtures
Iceberg tables in S3 via the AWS SDK: the loader writes, the connector reads, commits by conditional PUT; TLS, private CA and proxy for the object store	S3Mock (in-process and Docker), MinIO; racing writers on MinIO only; never real AWS S3
REST catalogs (OAuth2, ETag, vended credentials), incl. Glue's Iceberg endpoint and S3 Tables (SigV4)	REST protocol tests; `SigV4Test`; `GlueLiveTest` written, NOT RUN on AWS
Mixed storage per source system (`--store calypso=iceberg`)	`tools/lake/stores.py`; equality test: 18 answers the same
`IcebergLoader`, `IcebergMaintenance` on both engines; `load-iceberg.sh` and `.ps1` (Windows); orphan files, Admin → Lakes, the 7-year floor	`IcebergOrphanFilesTest`, `test_maintain_iceberg.py`
`deploy/aws/iceberg/`: load, backfill, maintain, lifecycle, alarms, report, IAM, Glue check; with the shared ECS images, EFS access points, Cloud Map, IAM roles and buckets	`--help` and `--dry-run` in `test_deploy_aws.py`; first real run pending
VaR, intraday and exposure from Iceberg; official results to Delta or Iceberg, disk or `s3://`	`VarFormatEqualityTest`, `ExposureFormatEqualityTest`: Delta = Iceberg to the cent; `LakeAppendsTest`; `OfficialS3Test`

S3 on three simulators, no Docker: what each test proves

Test	Simulator	What it proves	Runs on
`NativeOnS3SimulatorTest`	Adobe S3Mock 4.12.4 in the JVM; MinIO binary	the Iceberg loader writes, the connector reads a table in S3; writers at once where atomic	Linux, Windows (S3Mock)
`NativeOnS3Test`	S3Mock image; MinIO image (Testcontainers)	the same; the commit probe decides single-writer	where Docker is reachable
`S3MockContentStoreTest`	S3Mock in the JVM	the content store on S3	Linux, Windows
`tools/lake/test_s3_simulators.py`	moto 5.2.3 (pure Python); MinIO binary	**Delta** on S3; `lake publish`: new table, a day restated, a missing file refused, three at once	Linux, Windows (moto)
`OfficialS3Test`	S3Mock in the JVM	official results to `s3://` in Delta and Iceberg; two writers in sequence; the exclusive create	Linux, Windows
`check-conditional-writes.sh` (`delta-lake/`, `iceberg/`)	your real bucket	the bucket refuses a second create (412): the probe both formats rely on	AWS (first run pending)

Conditional writes honoured: S3Mock: create-only, update-if-match, delete-if-match, racing creates not atomic · moto: all three, racing creates one winner · MinIO: all, the most faithful (binary needed; skipped with the reason). Each suite probes and prints its line first.

```
./mvnw -q -pl drishti-identity,plugins/drishti-plugin-iceberg -am test -Dtest='S3MockContentStoreTest,NativeOnS3SimulatorTest' …
drishti-console/.venv/bin/python -m unittest -v tools/lake/test_s3_simulators.py      # moto: Delta and lake publish
```

Not yet proven on AWS: no table has been written to a real S3 bucket or committed through a real Glue catalog. First step of any production plan: `check-conditional-writes.sh`, then `check-glue.sh` against a development bucket (PROPOSED). Fixed: `lake publish --storage-option AWS_ENDPOINT_URL=…` checks the files on that endpoint, as the commit does.

The evidence: Delta = Iceberg, to the cent

Test	Same lines, Delta lake and Iceberg lake, one connector switched	Equal
`VarFormatEqualityTest`	VaR, ES, component and incremental VaR, worst scenarios, limits; every scope incl. the limit letters; USD and EUR; intraday file batches applied live; official records	to the cent
`ExposureFormatEqualityTest`	EE, PFE, EPE, profiles, contributions; official records	to the cent
`LakeAppendsTest`	the Kafka PV stream's window: one Iceberg append (on Delta, as before)	applied live
`IcebergParityTest`, `IcebergVectorsTest`	documents, vectors (both engines), identity, inventory against the Delta connector	bit for bit

On the demo data	Delta	Iceberg	Label
`VAR BANK`, 2026-09-30, USD	112,041,646	112,041,646	MEASURED (demo)
The risk domain moved (`--store risk=iceberg`)	14 tables, 955,124 rows	the same, in 579 s	MEASURED (demo)
`LL:LL-0003`, `XPO BANK`, official records	answered	answered from Iceberg	MEASURED (demo)

Demo data, not bank scale. The benchmark on both formats is **PLANNED**.

What 2.3 million positions put in the lake each day

Per business date	Arithmetic	Result	Basis
P&L vectors, float64 (in the lake)	2.3M × 509 × 8 B	9.37 GB[1]	arithmetic
... float32 (the hot copy)	2.3M × 509 × 4 B	4.68 GB	arithmetic
... as Parquet (≈ 0.75 × raw)	9.37 GB × 0.75	≈ 7.0 GB[2]	ESTIMATED
... a year; 7 years kept	× 252; × 7	≈ 1.77 TB; ≈ 12.4 TB	ESTIMATED
Partitions; files at 512–250 MB	1 (business_date); 7.02 GB ÷ 512–250 MB	1; ≈ 14–28 files[3]	PROPOSED
A cold read of the date	7.02 GB ÷ 1 MB row groups	≈ 7,000 GETs[4]	arithmetic
Portfolio definitions	15,000 rows a date	small	arithmetic
Every portfolio's vector, if stored	15k × 509 × 8 B	≈ 61 MB per currency[5]	PROPOSED
Kafka micro-batches	5-minute window	≤ 12 commits an hour	arithmetic
Intraday records on the topic	10,000 updates × 4–8 KB	40–80 MB a day	guide

The risk lake is small next to the documents

≈ 12.4 TB of VaR vectors over 7 years, against ≈ 200 TB of documents (slide 89). Exposure paths are the large table (slide 52).

Position grain

A system may send one row per position: rows = positions, so 2.3M rows a date whatever the trade count. 15,000 portfolios are far under the 100,000-row read limit.

[1] 2.3M: bank scale (product owner); 509: the scenario set [2] × 0.75 Parquet ratio (VAR_GUIDE.md 27.4); 252 dates a year, 1,764 in 7 [3] ÷ 512–250 MB files (VAR_GUIDE.md 29.2) [4] ÷ 1 MB row groups (LAKE_ROW_GROUP_KB) [5] 15,000 portfolios: bank scale Full derivations: slides 133–136.

Seven years at bank scale: Delta against Iceberg

	Delta (risk domain)	Iceberg (risk domain)
How 7 years are kept	log and removed files kept 2,557 days; checkpoints	keep every snapshot 7 years (`--expire-hours` ≥ 61,368[2])
Data kept, 2.3M positions	1,764 dates × 7.02 GB ≈ **12.4 TB**[1] of vector Parquet (+ replaced files of restated days)	the same ≈ **12.4 TB**: the data is format-neutral
Versions kept (full runs)	252 a year × 7 = **1,764** commits in the log	**1,764** snapshots listed in every metadata JSON
Intraday micro-batch commits	one small JSON per window per table	one snapshot per window per table (ASSUMPTION: the count depends on the window)
Cost per commit	a small JSON; replay from the last checkpoint	the metadata JSON rewritten with every snapshot kept
Measured	7,200 files list in 84 ms from a checkpoint	1,800 daily commits: 93 ms a commit, start 160 ms → 33 ms after regrouping
With intraday micro-batches	more small JSON files; checkpoints absorb them	more snapshots in every metadata file: compact intraday into the day's run
Guard against a short setting	`lake maintain` raises it (`retention-floor`)	`--floor-days 2557`; official tables: max snapshot age 7 years

The data volume is format-neutral; the history is not. PROPOSED for an Iceberg risk lake: keep full-run snapshots for seven years and compact intraday appends into the day's run (the content store keeps the official captures); the 7-year floor is built; measure the metadata JSON at 2,000+ snapshots (with the planned benchmark) before committing.

[1] 9.37 GB × 0.75 (Parquet against raw, VAR_GUIDE.md 27.4: 'about 70–80 % of raw') = 7.02 GB a date; × 252 business dates = 1.77 TB a year; × 1,764 dates (7 years) = 12.4 TB; ESTIMATED [2] 2,557 days (7 years) × 24 h = 61,368 h

Measured on the documents: a million trades a day, seven years

Question (1,000,000 trades a day)	Measured
type-ahead `TRD CLY-40834`	29 ms
open a trade, first time / cached	137–157 ms / 48 ms
open a trade on a past business day	288 ms
a search over the day, exact over 1,000,000	134–322 ms
desk P&L of all desks / another desk	327 ms / 48 ms
impact of a netting set (143,982 trades)	716–881 ms
a search on another business day, first / again	6.6 s / 306 ms
server heap in use	2–3.5 GB

Seven years: 1,800 days, 7,200 files	Measured
start-up incl. file list and newest ids	729 ms
a trade on 2019-12-05	22 ms
a trade on 2022-06-15	13 ms
the file list re-read	84 ms
an old day's columns	13 ms

MEASURED

Nothing in the read paths grows with history: a read concerns one business day, the file list comes from the last checkpoint, caches hold days by memory. A day in 2019 waits once for its id map (about a second).

Reading one VaR date at bank scale: lake, hot file, memory

The lake (S3, Parquet)

ESTIMATED

≈ 7.0 GB a date; a cold whole-bank pass ≈ 18–70 s on one server[1]

The hot file (local disk)

ESTIMATED

≈ 5.5 GB a date[2]; reopened in ≈ 0.5 s; 60 dates ≈ 328 GB

Memory (page cache + heap)

ESTIMATED

heap ≈ 1.8 GB a date[3]; whole bank ≈ 78 ms resident, ≈ 1.6 s cold from NVMe

× 23 = scaled from the measured 100,000-trade points (VAR_GUIDE.md 27.3); a node is size-independent. The 200K benchmark (measured on one workstation) keeps 2.3M extrapolated; VaR sharding (slide 68) is the step for a date larger than one server.

Measured at 100,000 trades	Time	× 23
first load of a date	0.8–1.2 s	≈ 18–28 s
reopen from its hot file	23 ms	≈ 0.53 s
a node's VaR and ES	0.48 ms	≈ 0.5 ms
the whole bank, float32	3.4 ms	≈ 78 ms

On ECS

`DRISHTI_VAR_HOT_DIR` and `DRISHTI_EXPOSURE_HOT_DIR` on fast local disk (NVMe), as the guides say: the default `<data dir>` is on EFS. Hot files are rebuildable from the lake, so they meet the NVMe exception of slide 22.

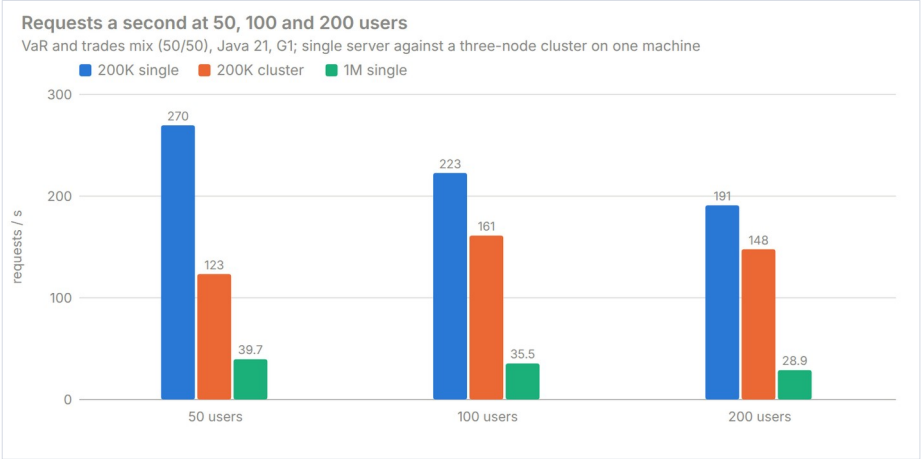
[1] the guide's 10–40 s per 4 GB × 7.02 ÷ 4 [2] 237.7 MB × 23 (measured at 100,000) [3] 80 MB × 23; 4.68 GB ÷ 60 GB/s; ÷ 3 GB/s NVMe (assumed) Full derivations: slides 133–136.

Measured: 200,000 and 1,000,000 positions side by side

Measure	200K, one server	1M, one server	200K, cluster of 3
requests a second, 50 / 100 / 200 users	270 / 223 / 191	39.7 / 35.5 / 28.9	123 / 161 / 148
p99 a VaR node, 50 / 200 users	0.72 / 5.39 s	1.41 / 26.0 s	0.46 / 5.08 s
p99 the bank minus a desk, 50 / 200 users	0.92 / 5.44 s	9.48 / 30.7 s	1.11 / 5.15 s
a date's first open	4.5 s	26.3 s	10.8 s
load: five kinds, two dates, Java loader	13.9 s	27.6 s	the same lake
heap after GC, two dates open	0.88 GB	7.5 GB	1.03 / 0.82 / 0.53 GB
resident memory (RSS) at the end	5.0 GB	21.2 GB	4.9 / 5.6 / 5.4 GB
disk a date: lake / hot file	0.29 / 0.56 GB	1.43 / 3.14 GB	the same lake

- **A cluster of three at 1,000,000 positions:** not measured on one machine[2]
- one server -Xmx8g at 200K, -Xmx16g at 1M; the cluster -Xmx4g a member, VaR sharded, 2 replicas a shard
- mix: half VaR questions, half trade pages; 60 s a step after a 45 s warm-up

Requests a second, by users[1]

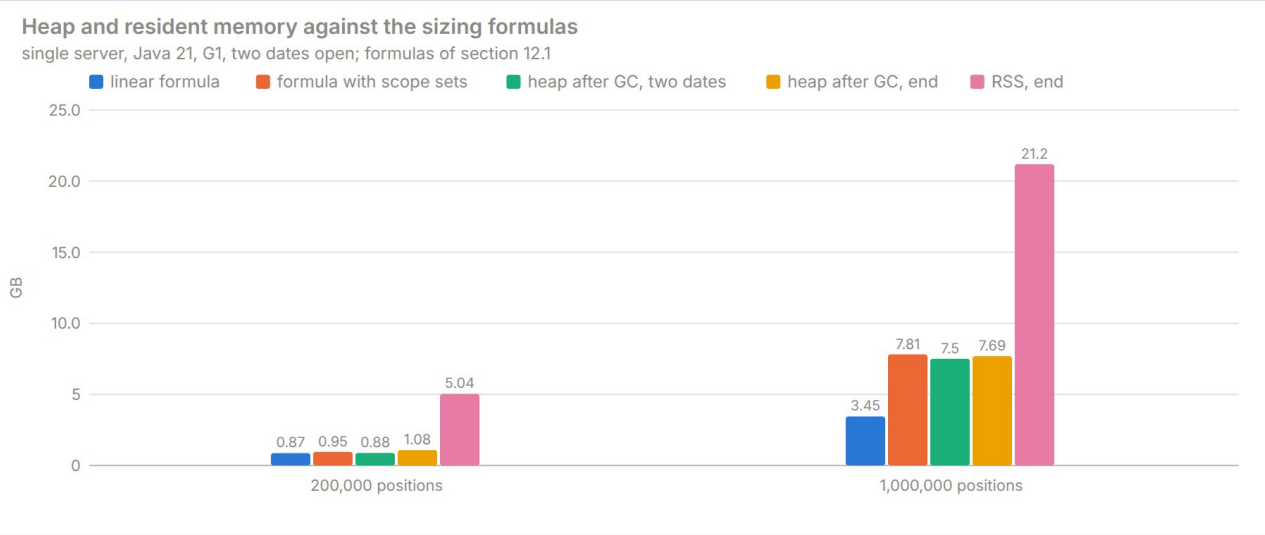


- 200K, one server: 270 a second at 50 users, every p99 under 1 s
- 1M: a node, a basket and contributions stay under 1.5 s at p99 with 50 users; the bank minus a desk sets the pace (two slides on)

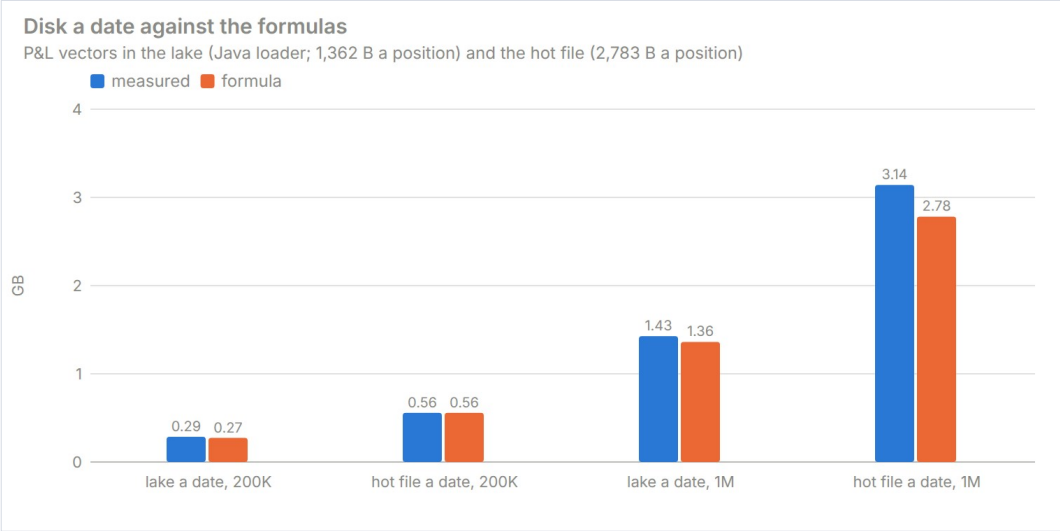
[1] measured on one workstation (12 cores, 24 threads, 61 GB, NVMe), Java 21, G1: BENCHMARK.md section 4 [2] three 16 GB servers exceed one 61 GB machine

Heap and disk against the formula, at both sizes

Heap and resident memory, two dates open



Disk a date: the lake and the hot file



The heap formula[1]

- `H = 128 MB + 980 B × P + 2 × (1,023 B × P + 23.6 KB × N + P × N ÷ 16)`
- P positions, N nodes, two dates open: 982 MB against 881 MB measured at 200K; 7.8 GB against 7.5 GB at 1M
- the P × N ÷ 16 term is the nodes' ordinal sets; -Xmx = (H + 0.13 GB) ÷ 0.61: 13 GB at 1M, run at 16 GB

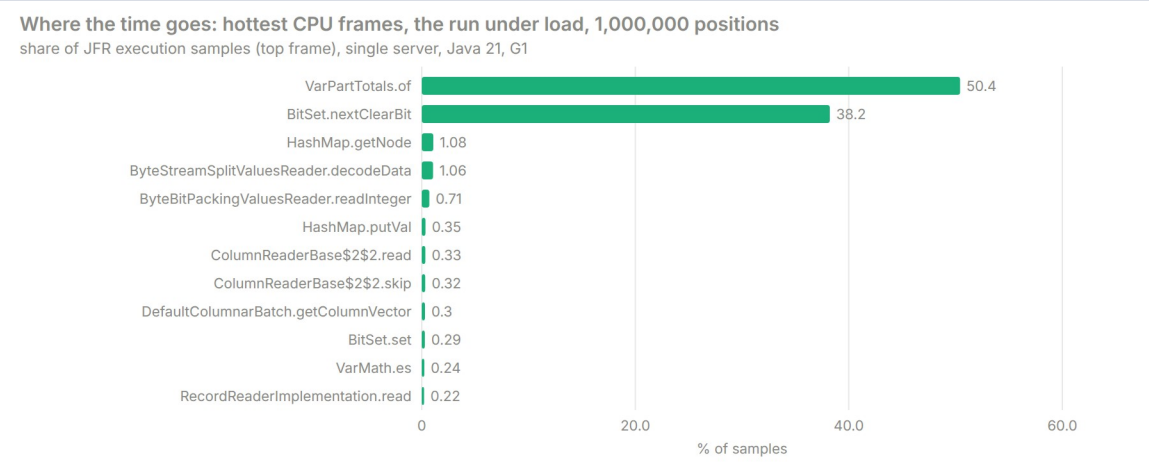
Disk a date

- the lake (S3): 1,362 B a position, 5% above it at both sizes
- the hot file (NVMe): 2,783 B a position, met at 200K, 13% above at 1M (639 MB of aggregates for 27,024 nodes)
- resident memory about three times the heap: the mapped hot files count in it

[1] fitted on both measured sizes, one server, Java 21, G1: BENCHMARK.md 4.8 and 7.1

Where the time goes and what we do next

Hottest CPU frames under load, 1,000,000 positions[1]



- summing a scope's parts position by position takes 89% of the CPU under load at 1M
- one user: a VaR node 3 ms at 200K, 4 ms at 1M; the bank minus a desk 68 ms and 1.46 s
- heap: 3.2 GB a date at 1M, mainly each node's ordinal set
- a first open: decoding the lake's Parquet is two thirds of its CPU

What we do next, in order

1 Sum a scope's parts by runs

- a position is one run of ordinals: the totals walk runs, not positions

2 The heap a date: a few hundred MB[2]

- the 3.2 GB comes mainly from the nodes' bit sets: compressed run sets and off-heap arrays

3 Ten dates in memory

- least recently used out first; month-, quarter- and year-ends pinned; a byte budget

4 Then Java 25, and BLAS later

- Java 25's accelerated paths over the hot files; later BLAS through OpenBLAS, for many scopes at once

[1] measured: JFR execution samples, one server, Java 21, G1 (BENCHMARK.md 4.9) [2] target, not yet measured (HIGH_PERFORMANCE_NATIVE.md section 3.3)

The parallel Java loader: JSON Lines into Delta or Iceberg

The pipeline (drishti-lakeload)

- read: a file in byte ranges, ranged GETs on `s3://`; `.gz` / `.zst` streamed
- parse: streaming Jackson, the vector straight to `double[]`
- sort and spill: runs to `memory.spill-dir`, bounded by `memory.budget-mb`
- write: Parquet files of `file-size-mb`; ONE commit per business date

The layout it writes

- sorted by `LEGAL_ENTITY`, `DESK`, `id`: files a shard owner can skip
- a file cut where a desk changes; a load manifest of each file's key range
- the vector column BYTE_STREAM_SPLIT + ZSTD; Delta through Kernel, Iceberg by replacePartitions
- folder or `s3://` in and out; credentials from the AWS chain only

```
drishti.py lake build --engine java --kind scenario-pv \
  --from s3://<landing>/<date>/pv/ --lake s3://<lake>/lake \
  --loader-set memory.spill-dir=/mnt/nvme/spill
```

Equal to the Python path on Delta and Iceberg, row for row and vector for vector (`PythonPathEqualityTest`). The Python path stays the default; `--engine auto --java-above-mb N` picks the loader for large loads.

Load	Time	Basis
10K positions × 510, Delta, warm	0.19–0.29 s	MEASURED
the same into Iceberg, warm	0.25–0.27 s	MEASURED
2.3M, 32 vCPU, 64 GB, NVMe spill	1–2 min[1]	ESTIMATED
2.3M, Fargate 16 vCPU, 120 GB	≈ 2 min	ESTIMATED
2.3M, 16 vCPU, 32 GB, gp3 spill	4–5 min	ESTIMATED

PARALLEL_LOADER.md section 11: 2.3M not yet measured

[1] PARALLEL_LOADER.md §11's estimate from about 92–150 µs of CPU a position (scaled from the 10,000-position ThroughputTest) plus spill I/O; to be confirmed by the planned benchmark

VaR sharded across servers, and running it on ECS Fargate

How it works

- each date cut into shards of `LEGAL_ENTITY`, `DESK` ranges (a desk never split)
- two owners per shard by rendezvous hashing; another zone with `zone-tag`
- an owner reads only its ranges from S3: Delta files skipped by the log's statistics, Iceberg's by their bounds
- every member answers every scope: nodes from aggregates, the rest by partial vectors

```
# config/packs/market-risk/pack.yaml
sharding:
  enabled: true
  key: [LEGAL_ENTITY, DESK]
  kinds: [scenario-pv, pnl-vector]
  target-shard-mb: 2048
  replicas: 2
```

On ECS Fargate: scale in safely

- stop hook: `drishti.py var shards --drain` (or `POST /api/v1/var/shards/drain`)
- wait for `GET /api/v1/var/shards` → `"safeToStop": true`
- `stopTimeout` ≥ the largest shard's build + `grace-seconds`

When it cannot answer exactly, it says so

- no live owner of a shard: `503 DRS-5215`, never a partial VaR
- another data version: `409 DRS-5216`; a bad cut: `422 DRS-5217`
- nodes (VAR BANK, desks) still answer from their aggregates

Equal to one server **to the cent** on three members, USD and EUR, Delta and Iceberg, intraday included (`VarShardClusterTest`). Off by default. Drain and rolling upgrades on ECS: chapter 6.

The demo lake: 85 Delta tables from committed JSON Lines

What	Value
Tables	**85**, about 3.4 GB
Business days	60, ending today's business date
Built in	943 s (about 15 minutes), measured
From	committed `data/testdata/…/*.jsonl.gz` (about 138 MB)
Trades	15,000 on the last day in 3 tables (ENDUR, CALYPSO, MUREX)
Risk	VaR inputs, limit letters, scenario PVs, FX scenarios, exposure paths for 74 netting sets

```
drishti.py lake demo                          # all 85 tables
build_demo_lakes.py --store risk=iceberg   # risk on Iceberg
build_demo_lakes.py --names config/demo/names.yaml
```

- **The pattern:** `data/testdata/<domain>/<kind>/*.jsonl.gz` → `build_demo_lakes` → Delta → the pack's Delta connector
- tested with the demo connector **off**: a build test fails any shipped kind without a Delta connector
- the same commands build a lake on EFS for a proof of concept

Domains: trading, reference, legal, collateral, credit, **risk**, market, macro, news, civic, us-equities, gleif, funds. **Not a production ETL:** in production your own loads end with the data-load signal (DATA_LOADS.md), and on AWS the loader task does the work (slide 74).

Public-data packs: real data on the same lake pattern

us-equities

- 479 companies with SEC filings
- filings, facts, index members, SIC
- prices internal only, off

gleif

- ≈ 9,500 legal entities
- 540 ownership groups, 132 countries
- LEI, BIC, ISIN mappings

funds

- 200 largest US funds and ETFs
- N-PORT holdings, N-CEN facts
- index weights from SEC data

refresh script

- `tools/publicdata/<pack>.py refresh`

→

committed snapshot

- `snapshot-<date>.jsonl.gz`

→

lake demo

- `<pack>/<kind>` every demo day

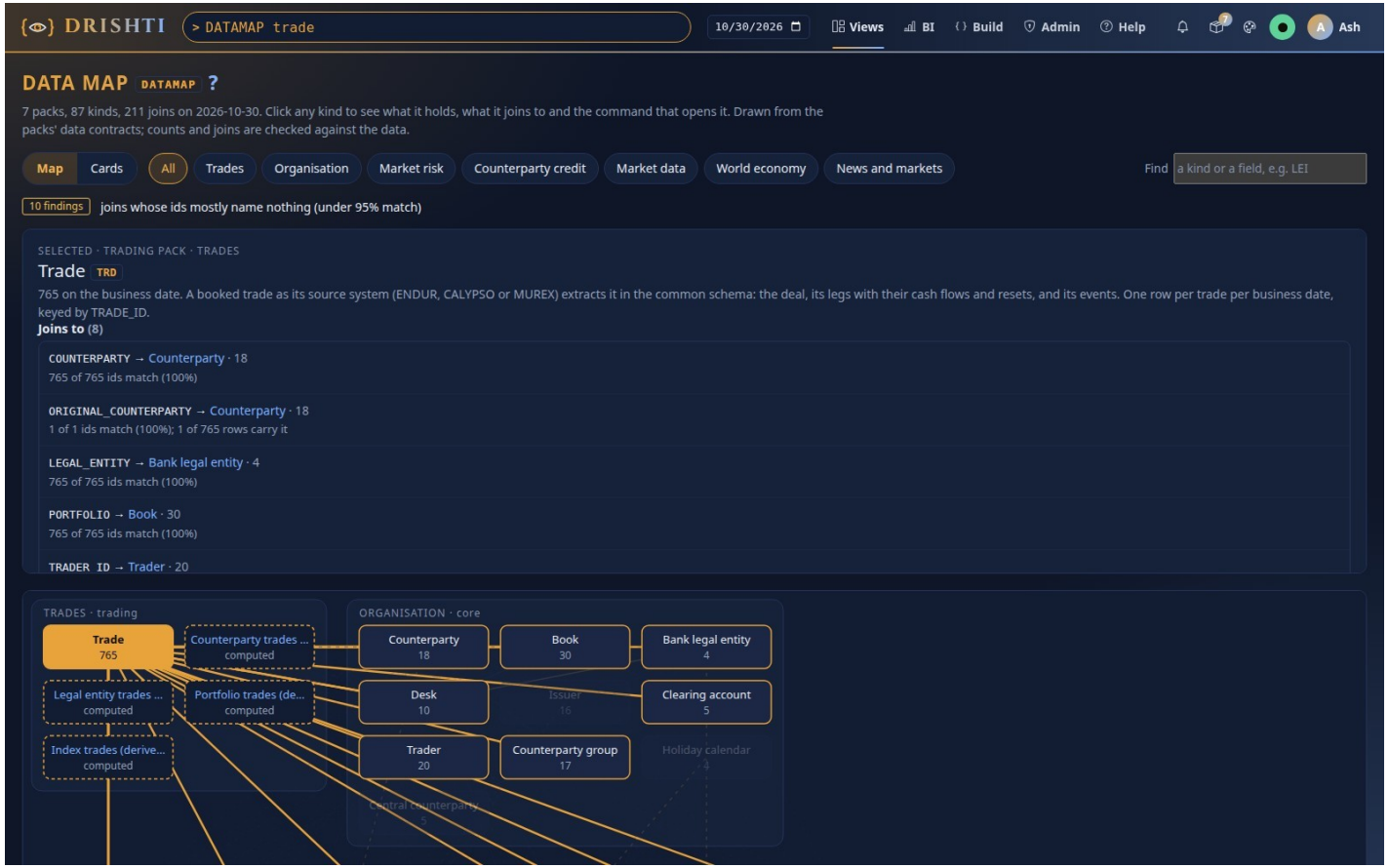
→

Delta connector

- `us-equities-store`, `gleif-store` …

- One standalone refresh script per pack (`status`, `refresh`, `--dry-run`, `--offline`), `.sh` and `.ps1` wrappers
- Internal-only sources (IMF, Yahoo, news) are refused in committed folders without `--internal-ok`
- Names in the type-ahead from promoted columns: `USEQ Ford` finds `F` (`layout.<kind>.type-ahead`)

DATAMAP: every kind, count and join, checked against the lake



`DATAMAP trade`: the selection across the top (Trade and its joins, each checked against the lake), the map at full width under it (QUICKSTART packs, 2026-10-30).

- kinds from the packs' contracts; computed kinds (VaR, exposure) dashed
- joins from `links:`, `link:` and `graph.fields`, **each checked against the lake**: under 95 % match is a finding
- counts per business date, cached 15 minutes; read from the day's columns, never documents

On the demo lake

Measured 2026-10-27, 10 packs: 96 kinds, 226 joins, 176 checked in about 1.5 s; 11 findings to fix in the contracts or the data.

Operating it

06



Setup at a glance: steps 0 to 11, Delta and Iceberg

0 Prerequisites

Delta: VPC, endpoints, EFS, KMS, ECS, ALB, SNS, secrets
Iceberg: + a Glue endpoint if Glue

1 Settings

Delta: `deploy/aws/env` from env.example
Iceberg: `ICEBERG_TABLES_FILE`, `ICEBERG_PREFIX`

2 Buckets

Delta: `setup-buckets.sh`
Iceberg: unchanged

3 IAM

Delta: `setup-iam.sh`
Iceberg: `iceberg/setup-iam.sh`, `iam/*.json`, Glue

4 Prove commits safe

Delta: `check-conditional-writes.sh`
Iceberg: the same check

5 Storage tiers

Delta: `setup-lifecycle.sh`
Iceberg: `iceberg/setup-lifecycle.sh`: `data/` only

6 EFS access points

Delta: `ecs/setup-efs.sh`
Iceberg: unchanged

7 Cloud Map seeds

Delta: `ecs/setup-cloud-map.sh` (twice)
Iceberg: unchanged

8 Images

Delta: docker build ×3, push to ECR
Iceberg: the server image carries `IcebergLoader`

9 Tasks, services

Delta: `ecs/register.sh`
Iceberg: + `DRISHTI_ICEBERG_ROOT`

10 First load

Delta: `load-day.sh --date …`
Iceberg: `iceberg/load-day.sh`; `check-glue.sh`

11 Alarms and report

Delta: `setup-alarms.sh`, `report.sh`
Iceberg: `iceberg/` versions: snapshots

Every script: bash and the AWS CLI v2, `deploy/aws/env` (git-ignored), `--help` and `--dry-run`, safe to run again; dry-run tested, not yet run in a real account. Step by step: appendix, slides 108 to 119.

The daily load on AWS: Delta's four steps, Iceberg's one

Delta (built): build locally, then publish

aws s3 sync

- landing → EFS scratch

↓

lake build

- scratch Delta table

↓

aws s3 sync

- data files → S3

↓

lake publish

- one commit, conditional PUT

Iceberg (built, `iceberg/load-day.sh`): the loader writes S3

IcebergLoader, engine native

- reads the landing dumps (stdin or a file)
- sorts each day by id, spills to EFS or ephemeral disk
- writes Parquet and manifests to `s3://…/iceberg/<domain>/<table>/`
- commits: conditional PUT of `vN+1.metadata.json`, then `version-hint.text`

```
deploy/aws/iceberg/load-day.sh --date 2026-10-20
  # per table in ICEBERG_TABLES_FILE, under its lock:
  # landing dumps → lake build --format iceberg \
  #   --lake s3://LAKE_BUCKET/iceberg
  # metrics: LoadSucceeded, LoadSeconds
```

Fewer moving parts on Iceberg: no scratch table and no second sync; the loader holds the table's lock only while it writes. Keep the Delta scripts' rules: one table per lock, `--dry-run`, exit codes, metrics to `Drishti/Lake`.

The data-load signal

lake publish
- one commit on S3

→

POST /api/v1/packs/{pack}/loads
- per date, a `loads:write` token

→

the server
- refresh, verify, alert (DATA_LOADS.md)

→

risk engines
- VaR: reload + official capture
- exposure: reload

- With `DRISHTI_SIGNAL_URL` (a node's Cloud Map name) and `DRISHTI_TOKEN_SECRET_ID` (an API token with `loads:write`, read into a 0600 file)
- Without it, servers see the new table version at their next refresh (`refresh-seconds`, 10)
- Either way readers switch at the commit; the signal makes it immediate and runs the load's expectations and alerts
- **Risk kinds:** a `pnl-vector`, `scenario-pv`, `scenario-set`, `risk-hierarchy` or `portfolio` signal reloads the VaR date and captures it officially; an intraday batch is applied on top; an `exposure-path` signal reloads exposure
- **Iceberg:** `lake publish` sends it for Delta; for Iceberg a script sends it (`build_demo_lakes.py --server` does so locally), else the servers see the new snapshot at the next refresh (10 s)

Signals are safe in bursts: reads queue at most `max-queued-reads` (4,096) and a read whose caller has gone leaves the queue.

Backfill and restatement, Delta and Iceberg


```
deploy/aws/delta-lake/backfill.sh --from 2026-01-02 --to 2026-10-16 --skip-missing          # weekdays; dates without dumps skipped
deploy/aws/delta-lake/backfill.sh --from 2025-01-02 --to 2025-12-31 --from-dir /opt/drishti/data/history
```

A source re-sent	Delta (`delta-lake/`)	Iceberg (`iceberg/`)
some whole days	`backfill.sh --dates 2026-10-14,2026-10-15`	load the days again: each replaces its partition in one commit
one product type of a day	`load-day.sh --date 2026-10-15 --tables trading/trade_endur --mode overwrite-partitions`	the whole day again (product type is not a partition)
years of history	`backfill.sh --from … --to …`, oldest first; stops at the first failing date and prints the command that resumes	`backfill.sh --from … --to …`, oldest first; stops and says how to resume
old days to drop	`lake maintain` retention	`lake maintain` retention (metadata only), 7-year floor for risk

- Re-running a date replaces it: a backfill can always be run again; a large one runs several loader tasks side by side on different tables (`--tables`), the per-table lock keeps two apart
- Readers switch at the commit, never a mixture; data files of an old restated day tier by their date prefix at the next lifecycle run (the rule is by date, not object age)

Restatement: *known at* before it still shows the day as it was, until vacuum (`LAKE_VACUUM_HOURS`; 7 years for risk) or snapshot expiry. A restated VaR or exposure date gets the next `RESULT_VERSION`; the earlier one is kept.

Guarded: removing more than half of a table's rows fails, removing nothing, unless `--force-drop`; a row dated after tomorrow in the business zone is not loaded and the load ends with an error naming it.

What the nightly maintenance runs, Delta and Iceberg

Nightly (`MAINTAIN_SCHEDULE`, Tuesday to Saturday 02:30) as the maintenance task, per table under the table's lock; then setup-lifecycle.sh.

```
drishti.py lake maintain --config <lake-maintenance.s3.yaml rendered> --once --force-foreign --tables trading/trade_endur
deploy/aws/iceberg/maintain.sh     # IcebergMaintenance per table; every option: appendix
```

Step	Delta (`maintain.sh`, lake-maintenance.s3.yaml)	Iceberg (`iceberg/maintain.sh` → `IcebergMaintenance`)
Retention	`keep-business-days` (`LAKE_KEEP_BUSINESS_DAYS`, 1800): dates older than that, one commit; refuses to drop more than `max-drop-share` (0.5)	`--keep-days 1800`: days older than seven years, metadata only; more than half the rows needs `--force-drop`
Checkpoint · manifests	`checkpoint: true`; removes commit files past `delta.logRetentionDuration`	`--rewrite-manifests`: one manifest per month; planning reads a few dozen
Vacuum · expiry, orphans	`vacuum-hours` (`LAKE_VACUUM_HOURS`, 168): files no version uses; S3 keeps them 14 more days as noncurrent	`--expire-hours 168`: keeps the metadata JSON small, bounds *known at*; `--remove-orphans`: files no snapshot names, 72 h at least, a dry run lists them
Compaction · relayout	`compact: false` (slide 79)	`--relayout`: re-sort days another writer appended out of order; fold delete files into data
Risk floor	7 years: 1,827 days, 61,368 h; `risk/`, `risk-results/` never below it (`retention-floor`)	`--floor-days 2557`: no date dropped, no snapshot expired inside 7 years

Iceberg, measured: connector start on 1,800 daily commits 160 ms → 33 ms after manifests and expiry; 93 ms a commit; drift check 0.7 s, keep + rewrite + manifests + expiry 2.9 s (10,000 trades).

> `--force-foreign`: the S3 Delta tables are Drishti's (lake publish wrote them), so the script overrides the ownership rule and marks the runs forced. A managed compaction must keep the id sort (PROPOSED rule).

Storage tiers by business date, for both formats

0 – 90 days	90 days – 2 years	2 years – kept days	beyond
S3 Standard	Standard-IA (`TIER_IA_AFTER_DAYS`)	Glacier Instant Retrieval	removed by `lake maintain` / `IcebergMaintenance` → noncurrent expiry

business date age → read latency: milliseconds in every tier

```
{"ID": "tier-trading-trade_endur-2026-03", "Status": "Enabled",
 "Filter": {"And": {"ObjectSizeGreaterThan": 131072,
   "Prefix": "lake/trading/trade_endur/business_date=2026-03"}},
 "Transitions": [
   {"Date": "2026-06-29T00:00:00Z", "StorageClass": "STANDARD_IA"},
   {"Date": "2028-03-30T00:00:00Z", "StorageClass": "GLACIER_IR"}]}
```

- **Never** Glacier Flexible Retrieval or Deep Archive: a restore of hours before a read
- **Never** an S3 expiry of current table files, in either format: it deletes files the log or a manifest still names; S3 keeps noncurrent versions 14 days

- One rule per table per business month, with dates (age is wrong for restated days); three months ahead; rewritten nightly
- **Iceberg:** the same rule, dates and tiers on `iceberg/<domain>/<table>/data/business_date=YYYY-MM`
- `_delta_log/` and `metadata/` never match; objects under 128 KB stay
- S3 allows 1,000 rules; the script refuses more (≈ 30 tables over 7 years[2]); each format's script keeps the other's rules

The risk lake: `risk/` and `risk-results/` keep 7 years of dates and time travel; old months sit in Glacier IR, still read in milliseconds, so any official figure can be recomputed years later.

Costs to know: Standard-IA keeps an object at least 30 days, Glacier IR 90 days; a restatement of a tiered day pays the rest of that minimum. Reads: Standard-IA ≈ USD 0.01/GB[1], Glacier IR ≈ USD 0.03/GB.

[1] S3 retrieval list prices (one Region) as quoted in AWS_DELTA_LAKE.md 9.4; minimum durations: AWS's rules [2] 1,000 rules ÷ about 32 rules a table (5 whole years in Glacier IR + 27 months moving) ≈ 31

Compaction on S3: relayout by id, compaction off by default

- Each day arrives laid out: files of `file-rows`, sorted by id, one commit ⇒ the S3 tables need no compaction
- A plain Delta compaction would merge files out of id order (single-entity reads would touch many row groups)
- `lake maintain` re-sorts (**relayout**) instead of compacting where the pack declares a layout
- So `compact: false` stays the shipped S3 setting; a day that needs re-laying is restated (slide 127)

Relayout, not compaction

`lake maintain` finds the layout of the connector whose table it is, so `trade_endur`, `trade_calypso` and `trade_murex` are re-laid out by id, not compacted out of order; relayout keeps `business_date` and `PRODUCT_TYPE` and checks each.

Turn compaction on for S3 after a relayout test there

PROPOSED

- the shipped S3 setting keeps `compact: false` (`lake-maintenance.s3.yaml`); turn it on only after this test
- run `lake maintain --dry-run` with `compact: true` on one S3 table, check id order per row group, then enable it nightly

Admin → Lakes: the same engine in the console

Step	What it does	Setting
Retention	deletes business dates older than the window, counted from today; refuses to drop more than half a table	`keep-business-days`
Compaction	merges small files, or re-lays a table in its pack's layout (dates that drifted only)	`target-file-mb` (128)
Checkpoint	readers replay a short log	`checkpoint` (on)
Vacuum	removes files no version uses, older than the window	`vacuum-hours` (168)

- **Dry run** first: files and size before → after, dates retention would drop, what vacuum would delete
- **Ownership rule:** only tables `lake build` built; a foreign table is read-only
- Schedules at a time or after every build; history kept in `maintenance-history.jsonl`
- **Iceberg tables** (`format: iceberg`, or auto under a local root) go to `IcebergMaintenance` under the same rules; the tables list has a Format column

On AWS: one scheduler

EventBridge runs `maintain.sh` nightly against `s3://` with `--force-foreign`.

Admin → Lakes maintains the lakes listed in `lakes.roots` and the file `<data>/lakes/lake-maintenance.yaml`; dry runs and history work the same.

Run schedules in **one** place: set `DRISHTI_LAKE_MAINTENANCE_SCHEDULES=false` on the console when EventBridge or the daemon runs them.

The risk floor applies in every place it runs.

What never scans the whole lake

Request	Reads	Grows with history?
Open one entity on a date	the snapshot or planned layout (in memory, refreshed every 10 s), the day's id map, one row group (≈ 1 MB)	no
History of one entity over N days	the days whose id map holds it; one read per day, eight at once	with N, not with the lake
Type-ahead	the newest day's ids and promoted names in memory	no
Search, pick list, derived kind, impact, DATAMAP count	that day's column set (id + promoted, never documents); files and row groups skipped by bounds	no
A VaR or exposure date	one streamed pass over the date's vectors, then the hot file	no
Refresh (Iceberg)	`version-hint.text` (path) or `loadTable` with ETag (REST); a changed snapshot's manifests	with live manifests: regroup monthly
Load, restatement, intraday batch	one day's files and one commit (Iceberg rewrites the metadata JSON)	no (Iceberg: metadata JSON grows with snapshots kept)
Retention	a delete by partition (the log or metadata only), one commit	no
Vacuum (weekly is enough)	a listing of the table's prefix	with the file count
Questions over all of history	not interactive: Athena, Trino or Spark on the same tables	—

The one Iceberg-specific growth: each commit rewrites a metadata JSON listing every snapshot kept: expire snapshots (seven-year risk tables: slide 61) and regroup manifests monthly.

History over the lake: TRD <id> 30

```
TRD 1000015 30                    # the trade over its last 30 business days with data
TRD 1000015 3M DIFF               # three months, each day's changes
GET /api/v1/history/trade/1000015?days=30&knownAt=2026-10-15T18:00:00Z    # as the lake stood then
```

days()

- the snapshot's `business_date=` partitions whose id map holds the id

one read a day

- id map → one row group; eight days at once

mask, then diff

- each day masked for the caller; changed fields

answer, cached

- by the kind's data version

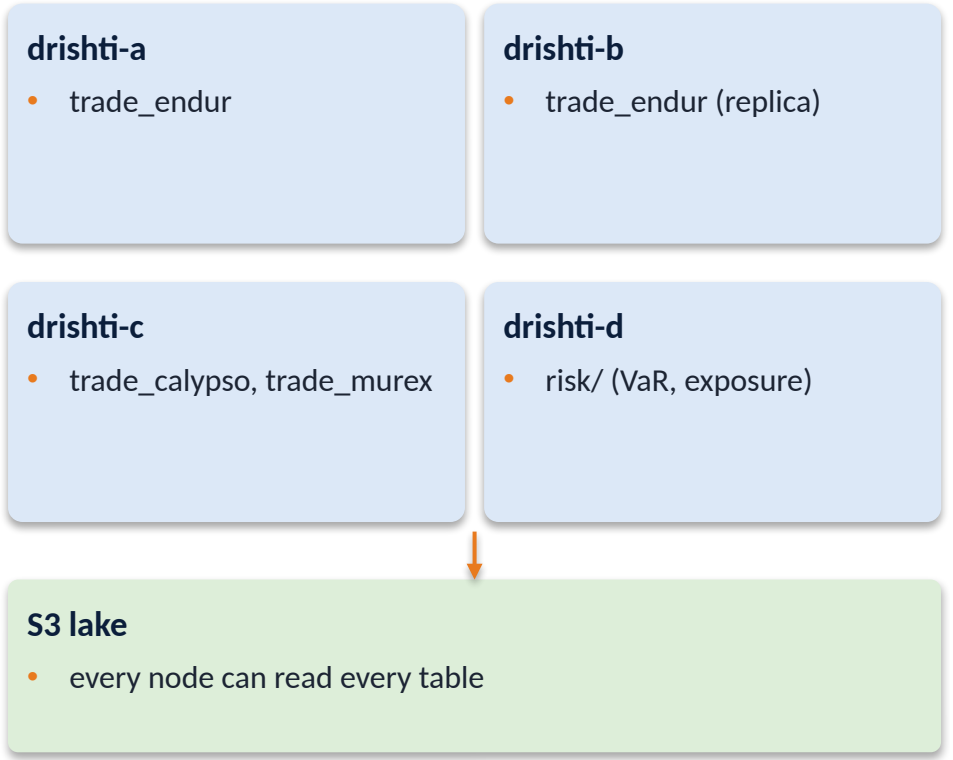
- A weekend, a holiday or a day the entity was not loaded costs nothing
- `knownAt` is Delta time travel: the day as the table stood then
- Across Drishti*net: one call to the peer that holds the data, masked on both sides
- Works the same on Iceberg (equality test) and on every connector (the contract's default)

Measured on 1,800 days

a trade on 2019-12-05 in 22 ms; the history reads only the days that hold the entity, so 7 years of lake cost no more than the days asked.

Spreading the work over Drishti*net

- **Shards:** nodes whose connectors read different tables of a kind; a search asks every one; a read goes to the holder
- **Replicas:** nodes on the same table; a forwarded read goes to the same node for the same entity (hash of its id)
- Every node can read every table from S3: a failed node costs speed, never answers
- Peer data over HTTP/2, one connection per peer; gossip on its own port; history across nodes
- **VaR sharded:** each owner reads only its legal-entity and desk ranges of a date (slide 68)
- **Iceberg tables too:** the plugin reports `iceberg:<table-uuid>`, so two nodes on one Iceberg table are replicas (a search asks one), also after a copy read through `relocate`



DESIGNED (SUPER_SCALE S5, S6):

ownership of (table, day) slices of the documents' tables on the ring, so N servers on one table hold N × one server's working set. (The VaR date is already sharded by legal entity and desk, partial vectors added in shard order: built.)

Zero downtime on ECS: readiness, drain, rolling upgrades

Drain
- POST /api/v1/admin/node/drain

→ **Safe to stop**
- safeToStop: true

→ **Update the service**
- the new task definition

→ **Services stable**
- aws ecs wait

→ **Ready**
- /actuator/health/readiness 200

Order: consoles → BI and data-serving → coordinators → shard owners, one zone at a time. Any failure stops the run, names the node and says how to resume (`--from`).

Readiness	HTTP	When
`UP`	200	joined, newest date open, shards ready
`DOWN`	503	joining, a date opening, a shard not ready
`OUT_OF_SERVICE`	503	drained before a stop

On ECS
- one service per node: `maximumPercent` 100, `minimumHealthyPercent` 0
- the task's health check is liveness: a node waiting for a date is never restarted
- the ALB target group reads the console's `/readyz`; owners are never behind it
- `stopTimeout` ≥ the largest shard's build + `grace-seconds`

```
export DRISHTI_DRAIN_TOKEN=drk_...      # an administrator's token with the node:drain scope
tools/rolling-upgrade.sh --inventory prod-ecs.yaml --version <version> --dry-run   # every step printed, nothing called
```

The inventory is `deploy/rolling.example.yaml` with `platform: ecs`: the cluster, the Region and each node's service and task definition.

Alerts to Prometheus: the platform rules, in PromQL

Alert	Expression (abridged)	Severity
`DrishtiServerDown`	up{job="drishti"} == 0, 2m	critical
`DrishtiCacheNearLimit`	cache bytes / limit > 0.95, 1h	warning
`DrishtiHeapAfterGcHigh`	live data / max > 0.85, 10m	warning
`DrishtiDirectMemoryHigh`	direct / max > 0.8	critical
`DrishtiDiskFreeUnderGuard`	free < guard: new dates refused	critical
`DrishtiSourceDown`	drishti_source_up == 0, 5m	critical
`DrishtiFeedsNeedAttention`	late, missing or failed loads	warning
`DrishtiViewsSlow`	page p99 above 150 ms, 5m	warning
`DrishtiMemberSuspect`	a member silent, 2m	warning

```
# prometheus.yml
rule_files:
  - drishti-alerts.yaml
scrape_configs:
  - job_name: drishti
    metrics_path: /actuator/prometheus
```

With or without Prometheus

- each server evaluates the same rules itself
- an alert that opens is logged, counted and sent to every administrator's inbox
- Admin → Health lists what is open now
- the CIO lens reads the same metric names

One file to load: `deploy/prometheus/drishti-alerts.yaml`; route `critical` and `warning` in Alertmanager. The lake's own CloudWatch alarms (step 11) stay as they are.

The numbers to watch: report.sh

```
bucket acme-drishti-prod-lake (CloudWatch daily storage metrics):
  StandardStorage                         10412.3 GB[1]
  StandardIAStorage                       46120.8 GB
  GlacierInstantRetrievalStorage         143998.1 GB
  objects                        672310

table                      version  checkpoint  behind   dates       newest   files GB(newest)    <32MB
trading/trade_endur           4512        4510       2    1764   2026-10-20     128      41.37        0
```

- `report.sh` is scheduled daily (`REPORT_SCHEDULE`), prints these and sends them to CloudWatch; numbers illustrative for ≈ 200 TB
- **behind** = commits since the last checkpoint: intraday risk tables add commits all day; checkpoints every 10 keep it short
- **<32MB** = small files of the newest day: should be 0 for the documents; intraday batches are small by nature (maintenance compacts them)

[1] illustrative output for a lake of about 200 TB (AWS_DELTA_LAKE.md 11.3): 10.4 + 46.1 + 144.0 TB = 200.5 TB, the 10 / 46 / 144 TB tier split of 12.3

The numbers to watch: healthy values and what to do

Number	Healthy	If not
behind (commits since the checkpoint)	under 10	a checkpoint failed: run maintain.sh; alarm log-long above 50
files of the newest day, and <32MB	≈ rows ÷ file-rows per partition; 0 small	tiny dumps or partitions too fine (slide 14)
live files per table	under ≈ 100,000	split the table (by source, or by year: S2, DESIGNED)
S3 5xx	rare	503 Slow Down on one prefix: spread with partitions; alarm lake-s3-5xx
Admin → Health p50Ms, p99Ms, cold reads	p95 < 300 ms newest days, < 1 s any day	footers and id maps not warm, or EFS latency
Admin → Caches: the one memory budget	under its cap	raise the heap (8 GB per million entities a day) or shard
Admin → Health `gossipPort`, `peerProtocol`	a port, `http2`	0: gossip on the server port (a firewall; slide 115)
LoadSeconds	well inside the window	one loader task per table, or larger loader tasks

The numbers to watch, for an Iceberg table

```
{"name": "trade-calypso", "health": "UP", "engine": "native",
 "cache": {"tables": 1, "snapshots": 1, "idMaps": 2, "columnSets": 2, "documents": 2,
           "partitions": 0, "daysWithDeletes": 0, "ids": 10000}}
```

Number	Healthy	If not	Status
`daysWithDeletes`	0 for Drishti's tables	single reads slow: `IcebergMaintenance --relayout`	
Health: `not laid out as the pack declares`	absent	`--columns trade:path,…` or reload	
`DEGRADED: … names files that are not there`	absent	a moved table: set the `relocate` pair Health gives	
snapshots kept (`IcebergSnapshots`, alarm at 500)	bounded by `--expire-hours`	commits and refreshes slow: expire	
live manifests per table	a few dozen	planning slow: `--rewrite-manifests`	**PROPOSED** metric
single-writer warning in the log	absent	the store ignores `If-None-Match`: one writer per table	

What 200 TB is (seven years of Parquet)

	Value
Per business day (252 a year)	about 113 GB[1] of Parquet; about 100 million documents at about 1.1 KB each compressed (Snappy)
Files per day	about 350 (330 MB each, 250,000 rows) over all tables
Live files after seven years	about 620,000[2] over all tables; three trade tables ≈ 200,000 each: above 100,000, so tables by year (S2) or more tables matter before year seven
Commits	one per table per day, plus maintenance: about 1,500 a year per table[3] (intraday risk tables more)
The risk lake at 2.3M positions	≈ 7 GB of VaR vectors (Parquet) a date, ≈ 12.4 TB over 7 years (slide 60)

ESTIMATED Estimates, to be replaced by the measurements of SUPER_SCALE step S8.

[1] 100 million documents × 1.1 KB (Snappy) = 110 GB, rounded up in AWS_DELTA_LAKE.md 12.1; 113 GB × 252 × 7 = 199 TB [2] 350 files a day × 1,764 dates = 617,400; 350 × 330 MB = 115.5 GB a day [3] the doc's estimate: 252 load commits plus nightly maintenance commits; not broken down

The deployment

Part	Size	Count
Server tasks	4 vCPU and 12 GB for the server (8 GB heap) in a 6 vCPU, 16 GB task with its console; 16 vCPU, 64 GB at SUPER_SCALE volumes	3 to 6, sharded by table
Loader task	4 vCPU, 30 GB, 21 GB ephemeral	one per table at load time
Maintenance and report tasks	the same image and size	nightly, minutes
EFS	each node's state and caches: 50–300 GB; loader scratch: one day of one table	one file system
A node reading the risk lake	heap ≈ 1.8 GB per open VaR date[1] at 2.3M; hot files ≈ 5.5 GB a date on local NVMe (EC2)	≥ 1; whole dates each

[1] 80 MB × 23 and 237.7 MB × 23, from the measured 100,000-trade VarTierBenchmark

Monthly cost: Delta, and what changes with Iceberg

Item	Basis	About (Delta)	With Iceberg
S3 storage, 200 TB tiered	10 TB Standard (USD 23/TB), 46 TB Standard-IA (12.5), 144 TB Glacier IR (4)	USD 1,400[1]	the same tiers; zstd may store less (not measured side by side)
S3 storage if all were Standard	200 TB × 23	USD 4,600	the same
S3 requests	250 reads a second, two GETs each, USD 0.40 per million; loads and lists	USD 500–700[2]	about the same; a few more small objects per commit
Retrievals from IA and Glacier IR	month-ends, audits, history	tens to a few hundred	the same
Versioning overhead	14 days of replaced and vacuumed files	a few % of storage	the same
Catalog	—	none	none (path-based); Glue or S3 Tables: their charges
Server nodes	3 tasks of 6 vCPU, 16 GB on Fargate, always on	USD 650–750[3]	the same tasks
Loader, maintenance	about an hour a day	USD 50–150	fewer steps; managed optimizers charged apart
EFS	1 TB Standard, Elastic throughput	USD 300 + throughput[4]	the same (the same disk caches)
CloudWatch, Secrets Manager, Cloud Map, endpoints, load balancer		USD 200–400	+ a Glue interface endpoint if Glue

The same 200 TB on EFS: USD 60,000 a month[5] in Standard; about USD 3,200 in IA plus USD 0.03 per GB read. **Either format, the same order of magnitude:** storage dominates; the choice is about capability and operations, not cost.

[1] tier split × S3 list prices [2] 250 reads/s × 2 GETs × a month × USD 0.40/M [3] 3 Fargate tasks × list prices × 730 h [4] 1,000 GB × USD 0.30 [5] 200,000 GB × USD 0.30 (EFS Standard) Full derivations: slides 133–136.

Disaster recovery, Delta and Iceberg

Event	Delta	Iceberg
A wrong commit	time travel: the previous version is readable until vacuum (7 years for risk); restate the days from the landing bucket (30 days)	*known at* reads the snapshot before it; restate the days from the landing bucket
Files deleted by mistake (a wrong vacuum, a script on the wrong prefix)	S3 versioning: restore noncurrent versions within `NONCURRENT_KEEP_DAYS` (14) (list-object-versions, copy-object)	the same
A table moved or copied (a laptop, another bucket)	—	`relocate: '<old> -> <new>'`, the pair Health gives (the library engine ignores it); the table identity stays
A Region lost	S3 Cross-Region Replication with RTC, EFS and ECR replication; fail over once the backlog is empty	CRR; read the replica with `relocate: 's3://prod/… -> s3://dr/…'` (PROPOSED: test it); repoint a REST catalog's metadata location too
A commit arrived before its files (replication)	that day fails (Health DEGRADED) until the file arrives	the same; a `version-hint.text` ahead of its metadata reads NoSuchTable until it arrives
Node state lost	AWS Backup of EFS; caches and hot files rebuild from the lake	the same
Cluster state (content store)	s3 backend: the bucket's versioning and replication; filesystem on EFS: AWS Backup and EFS replication	the same

The Iceberg-specific DR rule: absolute paths make a copy a different table until `relocate` maps it; with a REST catalog, the catalog's metadata location must be repointed in the recovery Region too.

Three machines: on premises and AWS

07



Three machines: the smallest highly available Drishti

Load balancer: a pair + VIP, or an ALB · routes on `GET /readyz` · no stickiness

Machine 1 · zone a

- console, 4 workers
- server: owner and coordinator
- about 2/3 of the shards
- every node's aggregate
- hot files on NVMe or EBS

Machine 2 · zone b

- console, 4 workers
- server: owner and coordinator
- about 2/3 of the shards
- every node's aggregate
- hot files on NVMe or EBS

Machine 3 · zone c

- console, 4 workers
- server: owner and coordinator
- about 2/3 of the shards
- every node's aggregate
- hot files on NVMe or EBS

Drishti*net: gossip + signed member calls between the three

The lake, shared

- S3 on AWS
- MinIO or the bank's S3-compatible store
- or NAS/SAN at one path

PostgreSQL, highly available

- RDS Multi-AZ on AWS
- primary + standby (Patroni)
- never the SQLite default

Why three

- **One fails:** every shard still has a replica; the survivors rebuild the second copy from the lake
- **One in maintenance:** two copies of every shard remain on the other two
- **Each holds about 2/3** of the shards (2 replicas ÷ 3), on average

Why not two

- fine for a pilot or UAT
- a failure or an upgrade leaves one copy of everything, and no machine to rebuild a second

Not shared: each machine's hot files and caches, rebuilt from the lake when lost. Secrets (token, net, session, embed key) the same on all three.

Three machines: failover and maintenance

One machine fails

- the other holder of each shard answers at once
- after 35–40 s the survivors rebuild the second copies from the lake
- its console drops out of the balancer; pages reload on another

A rolling upgrade

- drain one machine: its shards' next holders build them
- safe to stop → upgrade → start → ready
- then the next; two copies of every shard throughout

A network split

- both sides read the same lake at the same versions: read answers identical
- a shard held only across the split: refused (DRS-5215), never a partial answer
- writes go to the one shared PostgreSQL primary

Readiness	HTTP	When
`UP`	200	joined, newest dates open, owned shards ready
`DOWN`	503	joining, a date opening, a shard not ready
`OUT_OF_SERVICE`	503	drained before a stop

No consensus protocol

- ownership: a function of the shard map and the gossip view
- disagreement costs a refused call, never a wrong number
- every member keeps every node's aggregate

Still to come

PLANNED

- singleton jobs (official captures, the PV stream reader) by a lease in the shared database
- a tested PostgreSQL standby failover guide for on premises (RDS Multi-AZ fails over by itself in about a minute)

Three machines: sizing a machine, 200K to 10M positions

A machine of three	200K	1M	2M	10M
heap after GC, two dates open[1]	1.5 GB	3.1 GB	5.2 GB	21.6 GB
-Xmx: need → setting	2.7 → 8g	5.4 → 8g	8.7 → 12g	35.6: over 32 GB
hot file a date, whole bank	0.56 GB	2.8 GB	5.6 GB	27.8 GB
RAM: need → provisioned[2]	10 → 32 GB	19 → 32 GB	29 → 48 GB	111 → 192 GB
local NVMe: need → provisioned	23 GB → 200 GB	116 GB → 200 GB	232 GB → 500 GB	1,159 GB → 2 TB
full width: -Xmx need	12.0 → 16g	20.7 → 24g	31.5: at the line	118: needs D1
full width: RAM need → provisioned	21 → 48 GB	55 → 96 GB	97 → 192 GB	435: not three machines
lake a year, shared (measured / design)	0.07 / 0.15 TB	0.34 / 0.77 TB	0.69 / 1.54 TB	3.4 / 7.7 TB

ESTIMATED

Reading it

- **128 GB, 1–2 TB NVMe** carry 1-day VaR to 2M and every horizon to about 1M
- 2M with every horizon: 192 GB
- 10M: 1-day on 192 GB machines; every horizon needs a larger estate

Keep -Xmx below 32 GB

- with G1 the JVM compresses pointers below it
- the heap is not divided by members: each holds the whole skeleton

[1] H = 0.323 + 2 × (1,023 B × P + 19.5 KB × N), measured at 200K; N = 20,000 (ASSUMPTION) [2] heap + page cache + a load + 7 GB, ÷ 0.61; full width: the capacity model × P ÷ 2.3M; EXTRAPOLATED, to be confirmed at 1M Full derivations: slides 133–136.

Three machines on premises: the bill of materials

Item	Count	Size	Note
Drishti machines (server + console)	3	24–32 cores, 128 GB RAM, 1–2 TB NVMe; 192–256 GB for 2M with every horizon	one per rack or hall (`zone-tag`)
Shared object storage (the lake)	1 cluster	MinIO or the bank's S3-compatible store, erasure-coded over 4+ nodes, 2–4 TB usable to start	or NAS/SAN at one path on every machine
PostgreSQL	2	primary + standby: 8 cores, 32 GB, 500 GB SSD	streaming replication; Patroni or the bank's operator
Load balancer	a pair	HAProxy + keepalived VIP, or the bank's F5	TLS, `GET /readyz`, no stickiness
Network	—	10 GbE between the machines and to the store	member calls and gossip inside the app zone
Kafka	optional	the bank's brokers	only for intraday PV streaming

The lake is small: at 1–2 million positions with 1-day vectors, about 1–3 TB covers several years of history, not tens of TB; it grows with positions, horizons and years kept.

Three machines on AWS: the reference

Piece	Choice	Note
Drishti machines	3 × **m7i.8xlarge**: 32 vCPU, 128 GB, one per zone	`r7i.8xlarge` (256 GB) for 2M with every horizon
Hot files	EBS **gp3**, 1 TB a machine, 6,000 IOPS, 500 MiB/s	or NVMe: `m8id.8xlarge`, 1.9 TB
The lake	**S3**, gateway endpoint	Delta with conditional writes, or Iceberg
Database	RDS PostgreSQL **Multi-AZ**	fails over in about a minute
Load balancer	**ALB** across three zones	target group on `/readyz`
Cheaper option	**Graviton** `m7g.8xlarge`, `r7g.8xlarge`	after confirming ARM builds of RocksDB, DuckDB, zstd, lz4

Are machines this large on AWS?

- **Yes, with room:** m7i runs to 192 vCPU and 768 GB in one instance
- r7i (memory-optimised) to 1.5 TB
- one gp3 volume: up to 64 TiB, 80,000 IOPS, 2,000 MiB/s
- the reference uses mid sizes: an 8xlarge has 10 Gbit/s to EBS

Monthly cost on the next slide: the same price constants as the VaR engine deck (Canada (Central), on demand), with 1-year and 3-year Savings Plans.

Three machines on AWS: the monthly cost

Item	USD a month
3 × m7i.8xlarge, on demand[1]	3,937
EBS gp3, 3 × 1 TB	363
S3 lake, 2 TB Standard	60
RDS PostgreSQL Multi-AZ (ASSUMPTION)[2]	600–1,300
ALB, data between zones, loads	82
NAT, endpoints, CloudWatch, secrets	395
Total, on demand	**5,400–6,100**
1-year Compute Savings Plan[3]	4,400–5,100
3-year Compute Savings Plan	3,500–4,200

ESTIMATED

Region and prices

- Canada (Central), on-demand list prices, 730 h a month
- the same price constants as the VaR engine deck's AWS cost
- **check every price in the AWS Pricing Calculator before quoting**

Not included

- consoles (they run on the three machines)
- Kafka, a disaster-recovery copy, support, taxes
- Graviton would save about USD 750 a month (ASSUMPTION)

[1] half of m7i.16xlarge USD 3.5952/h × 3 × 730 h (Canada (Central)) [2] not in the price constants: us-east-1 list + ~10% [3] EC2 −26.9% / −49.5%, the constants' rates, no upfront Full derivations: slides 133–136.

Fast disks on AWS

Disk	Speed	Survives	Price, approximate[1]	For Drishti
Instance-store NVMe: `m8id`, `m7gd`, `r7gd`, `i4i`, `i7ie`	~100 µs; 100Ks to millions of IOPS	a reboot; **not** stop or replacement	in the instance price	**hot files** (a rebuildable cache)
EBS io2 Block Express	256,000 IOPS, 4,000 MB/s, sub-ms	stop and start; 99.999%	$0.125/GB + $0.065/IOPS-month	not for a cache: ~7 × gp3
EBS gp3	3,000 IOPS, 125 MB/s included; to 80,000, 2,000 MiB/s	stop and start	$0.088/GB + extra IOPS, MiB/s	hot files that must survive a stop; RDS
FSx for Lustre	shared; hundreds of GB/s	persistent or scratch	~$0.15–0.60/GB	not needed
S3 Express One Zone	single-digit ms reads	one zone only	~$0.11/GB	optional: the newest dates
S3 Standard	10–100 ms first byte; wide	three zones	$0.025/GB	**the lake**

Recommendation

- hot files on **instance-store NVMe**: a replaced instance rebuilds each date from the lake on its first open
- or **gp3** with extra IOPS if they must survive stop and start; io2 is not worth it for a cache
- the lake on **S3 Standard**, optionally S3 Express One Zone for the newest dates; RDS on gp3

The NVMe variant of the reference

ESTIMATED

- 3 × **m8id.8xlarge**: 32 vCPU, 128 GB, 1.9 TB NVMe
- about USD 800 a month more than 3 × m7i.8xlarge with gp3[2]

[1] gp3 and S3 Standard: the price constants (Canada (Central)); io2, FSx for Lustre, S3 Express One Zone: us-east-1 list prices; ~7 × gp3: 1 TB at 16,000 IOPS, io2 USD 1,165 against gp3 160 a month; ESTIMATED, check in the AWS Pricing Calculator [2] m8id.8xlarge = m7i.8xlarge 1.7976/h × (us-east-1 2.0883 ÷ 1.6128) = 2.3276/h; 3 × 730 h − 3 × m7i − 3 TB gp3 = USD 798; ASSUMPTION, not in the price constants

EC2, ECS on EC2 or Fargate

	Plain EC2	ECS, EC2 capacity / Managed Instances	ECS on Fargate
Hosts	yours	yours; or AWS-managed for a fee (~12%)[1]	none
Largest member	any instance	any instance, one task a host	**32 vCPU; 60, 120 or 244 GiB**[2]
Local disk	instance-store NVMe	instance-store NVMe in the task	no NVMe; 200 GiB ephemeral; EBS at launch
Stop time	systemd, as long as needed	the agent's stop timeout	**at most 120 s**: the 90 s drain fits
A replacement	same disk	a new host's NVMe is empty	always a cold member; handover automatic
Always-on price	the instance	the instance (+ fee)	higher: 32 vCPU, 120 GiB $2.01/h vs $1.80[3]

Recommendation

- **VaR members**: ECS with EC2 capacity (or Managed Instances), one task a host, NVMe instances
- **consoles**, scheduled loads and maintenance: Fargate
- three machines: plain EC2 with the systemd unit is as good and simpler

A Fargate member?

- by memory a 32 vCPU, 244 GiB task carries a member of three: 1-day VaR to 10M, every horizon to 2M
- only with hot files on an EBS volume (200 GiB ephemeral: to about 1M)
- every replacement is cold; a 16-vCPU, 128 GB owner of the larger estate needs a 32-vCPU task

[1] ECS Managed Instances fee 0.431424/h on m7i.16xlarge at 3.5952/h (price constants) [2] AWS: 32-vCPU Fargate tasks announced mid-2026 (aws.amazon.com/about-aws/whats-new/2026/06/amazon-ecs-fargate-32vcpu); ECS developer guide, Fargate task sizes [3] Fargate 32 × 0.04456 + 120 × 0.004865 a vCPU- and GB-hour; m7i.8xlarge half of m7i.16xlarge; Canada (Central), on demand; 244 GiB: $2.61/h

What comes next

08

Designed: the super-scale scale-out of the lake

Item	What it adds	Why it matters at scale
S1	Range maps from the log's per-file id statistics; Parquet Bloom filters on id	find an id without building a day's id map
S2	Tables by year (`root: …/{year}`), closing a year	a table's checkpoint stops growing with seven years of live files
S3	A loader fleet writing slices of one table at once; a day register	loads of very large days finish in the window
S4	Products (link indexes, group totals, rollups, id lists) written with the data	aggregates and links without scanning
S5, S6	Each server owns a share of (table, day) slices on the ring	N servers hold N × one server's working set
S7, S8	Warm tier of memory-mapped column files; benchmark at scale in AWS	hot days from memory; measured numbers

Built, so not in this table: VaR sharded across servers (each owner holds only its ranges of a date; partial vectors added in shard order) and Drishti*net shards by table.

DESIGNED

Proposed by this deck: not yet in the docs

#	Proposal	Why
1	Set node.yaml `gossip-port` to a fixed number on ECS and open it in the `drishti-node` security group	`auto` changes the port at each start; behind a security group gossip falls back to 18480
2	Kafka windows written to an EFS lake, then copied to S3 by `lake publish --mode append` after each window	the stream appends to a lake path, not `s3://`
3	Re-enable `compact` for the S3 trade tables after a relayout test on S3	`lake maintain` now lays out per-source tables
4	`exposure-path` partitioned by date and run, sorted by netting set (as EXPOSURE_GUIDE.md proposes)	a netting set in one streaming pass
5	A published table of node and portfolio vectors per currency (from the VaR deck)	BI and peers read sums, not 2.3M rows
6	Change data feed on the risk input tables; Z-order or liquid clustering by book and portfolio (VaR deck)	followers see intraday deltas; a desk's read skips files
7	Load the risk lake with the parallel Java loader (`lake build --engine java`), not the Python path	its files are sorted by legal entity and desk, BSS + ZSTD: a node is a run of rows
8	Iceberg: path-based tables first; Glue's REST endpoint when Athena or Spark must share a table; S3 Tables after a proof	the same commit safety as Delta with no extra service (slide 37)
9	Iceberg format v3 reading (deletion vectors, new types), named branches and tags; a *Make a pack* button for `iceberg-external`	listed as later in the Iceberg docs

PROPOSED

Limits and what comes next

Limit	Status	Source
Not yet run in a real AWS account (steps 0–11, runbooks); no Iceberg table written to a real bucket	to do	AWS_DELTA_LAKE.md §15
The Glue and S3 Tables live runs (code and `check-glue.sh` written)	NOT RUN	AWS_DEPLOYMENT.md §22
The benchmark: 200K positions, one server against three sharded, both formats; 2.3M extrapolated; demo figures are demo only	PLANNED	IMPLEMENTATION_PLAN.md
Delta in-place schema evolution; Delta ↔ Iceberg conversion of any table; Unity Catalog	PLANNED	IMPLEMENTATION_PLAN.md
Kafka windows are appended to a lake path (local or EFS), not `s3://`	limit; to S3 PROPOSED	VAR_GUIDE.md §8.4
More than 100,000 hierarchy nodes or portfolios from a connector that cannot list a date whole (15,000 portfolios fit)	limit	VAR_GUIDE.md §20
Tables by year, range maps, slice ownership (S1–S8: scale-out)	DESIGNED	SUPER_SCALE.md
`lake build` sorts a partition only when it fits a batch	limit	AWS_DELTA_LAKE.md §7.2
A bucket without conditional writes: one writer per table	limit	AWS_DELTA_LAKE.md §8.2
Iceberg spec gaps, refused by name: format v3 (deletion vectors, new types), encrypted tables, ORC and Avro data files, branches and tags	limit	DRISHTI_ICEBERG.md §5
The native Iceberg engine reaches local disk and S3 only; Azure, GCS, HDFS need `engine: library` (Hadoop)	limit	ICEBERG_CONNECTOR.md §17
Doc fix: ICEBERG_CONNECTOR.md §4 says path tables on S3 'need a single writer' (true only of the library engine)	to do	ICEBERG_CONNECTOR.md §4

Decisions to take, with this deck's recommendation

#	Decision or action	Recommendation
1	Disk caches and VaR/exposure hot files on local NVMe (breaks the identical-path rule) or EFS	Hot files on NVMe from the start (the guides ask for fast local disk); connector caches on EFS until p99Ms shows EFS latency
2	First real run in a development AWS account	Before any production commitment: steps 0–11 and the runbooks; `check-conditional-writes.sh`, `check-glue.sh` (Glue and `--s3-tables`), one Iceberg load and read; record the outputs in the guide
3	Official results straight to `s3://`; how Kafka windows reach S3	Official: `s3://` root, add `s3:PutObject` to the server role. Kafka: EFS lake, then `lake publish --mode append`, 7-year retention (PROPOSED)
4	A fixed gossip port on ECS	A number in node.yaml and a security-group rule node to node (PROPOSED)
5	Compaction on S3	Re-enable after a relayout test on one S3 table (PROPOSED)
6	Tables by year (S2); range maps, Bloom filters (S1); slice ownership (S5, S6); VaR sharding on	When live files near 100,000 per table, or one server's working set or one VaR date becomes the limit
7	Which format for the **risk domain** (vectors, official results)	Delta by default; Iceberg where the bank standardises on it (equal to the cent); decide on a bank-scale benchmark's measured figures
8	Which format for **trades and reference data**	Either, per source system; Iceberg where the bank's pipelines already write it
9	The **catalog** on AWS for Iceberg	Path-based first; Glue's REST endpoint when Athena or Spark must share the tables; S3 Tables only after a proof
10	Seven-year history on Iceberg	The floor is built; keep full-run snapshots, compact intraday; measure metadata at 2,000+ snapshots

Appendix



Step 0: prerequisites (the platform team's templates)

- A VPC with private subnets in three Availability Zones
- A gateway VPC endpoint for S3 on their route tables (lake traffic never leaves the VPC; the bucket policy requires it)
- Interface endpoints for ECR, CloudWatch Logs, Secrets Manager and STS
- An EFS file system: Regional, encrypted, Elastic throughput, a mount target per subnet
- A KMS key for the lake; its key policy allows the roles of step 3
- An ECS cluster
- An Application Load Balancer with a target group for the consoles on port 17480
- An SNS topic for alarms
- The four secrets of AWS_DEPLOYMENT.md §11.3 (token secret, session secret, cluster key, admin seed)

Who builds these?

The platform team, with its own CloudFormation or Terraform templates. The Drishti scripts take their ids (subnets, file system, key ARN, cluster, target group, topic) from `deploy/aws/env`.

Run the scripts from

a workstation or admin host with AWS CLI v2, the repository, Java 21 and `drishti-console/.venv`.

Content store on S3 (optional): its bucket needs conditional writes too (step 4).

Step 1: the settings file


```
cp deploy/aws/env.example deploy/aws/env        # git-ignored; holds account ids and names, never a secret
$EDITOR deploy/aws/env                          # replace every <placeholder>
cp deploy/aws/delta-lake/tables.example deploy/aws/delta-lake/tables.conf   # your tables; set LAKE_TABLES_FILE
```

You should see: nothing yet. A script run with a placeholder left names it: `setup-buckets.sh: LAKE_KMS_KEY_ARN is still a placeholder (<arn:aws:kms:…>) in deploy/aws/env`

```
trading/trade_endur     trade   config/packs/trading   trade-endur     {date}/endur/          -      PRODUCT_TYPE
```

Field	Example	Meaning
folder	`trading/trade_endur`	the table's folder under the lake prefix (domain / table)
kind	`trade`	the Drishti kind
pack	`config/packs/trading`	the pack whose layout (key, promoted and vector columns) is used
connector	`trade-endur`	the pack's connector that reads this table
landing prefix	`{date}/endur/`	where the JSON Lines dumps are; `{date}` is the business date
business-date field	`-`	the field holding the date, or - for the day being loaded
partition column	`PRODUCT_TYPE`	the second partition key

Step 2: the buckets


```
deploy/aws/delta-lake/setup-buckets.sh --dry-run
deploy/aws/delta-lake/setup-buckets.sh
+ aws s3api create-bucket --bucket acme-drishti-prod-lake --object-ownership BucketOwnerEnforced \
      --create-bucket-configuration LocationConstraint=eu-west-2
+ aws s3api put-bucket-versioning --bucket acme-drishti-prod-lake --versioning-configuration Status=Enabled
+ aws s3api put-bucket-encryption --bucket acme-drishti-prod-lake --server-side-encryption-configuration file://…
+ aws s3api put-bucket-policy --bucket acme-drishti-prod-lake --policy file://…/policy-….json
setup-buckets.sh: done: acme-drishti-prod-lake, acme-drishti-prod-landing, acme-drishti-prod-logs
```

- **Three buckets:** `-lake` (every Delta table, the risk lake included), `-landing` (JSON Lines dumps, 30 days), `-logs` (access logs, 400 days)
- **On the lake:** versioning on; SSE-KMS with S3 Bucket Keys; Block Public Access; bucket-owner-enforced ownership; TLS only; VPC endpoint only (plus `drishti-<env>-admin*` roles); access logs; request metrics on `lake/`
- **Safety net:** noncurrent versions expire after 14 days; incomplete multipart uploads abort after 2 days
- **Idempotent:** run again, it says `acme-drishti-prod-lake exists: kept, settings applied again`
- `--dry-run` prints each changing call as `+ aws …` and each read as `? aws …`; none is made

Step 3: who may do what (IAM)


```
deploy/aws/delta-lake/setup-iam.sh --dry-run
deploy/aws/delta-lake/setup-iam.sh
```

Role	Policy (deploy/aws/delta-lake/iam/)	May
`drishti-<env>-server-task`	`server-policy.json`	read the lake prefix (GET, LIST), decrypt; mount EFS through access points
`drishti-<env>-loader-task`	`loader-policy.json`	read the landing bucket; write data files and commits; never delete (an explicit deny)
`drishti-<env>-maintenance-task`	`maintenance-policy.json`	checkpoints, retention and vacuum: the only role that deletes; the lifecycle rules
`drishti-<env>-execution`	AmazonECSTaskExecutionRolePolicy + `drishti/<env>/*` secrets	ECS pulls images, writes logs, injects secrets
`drishti-<env>-scheduler`	inline	EventBridge Scheduler runs the lake tasks

You should see `setup-iam.sh: done: three policies, five roles (…)`. The shipped server policy only reads; for official results to S3 add `s3:PutObject` on their prefix, no delete (AWS_DEPLOYMENT.md §22; slide 49).

Step 4: prove that commits are safe

```
deploy/aws/delta-lake/check-conditional-writes.sh
check-conditional-writes.sh: ok: the second create was refused (412 PreconditionFailed): concurrent commits are safe
```

- **What it does:** creates a test object with `If-None-Match: *`, then tries again; S3 must refuse the second with 412
- **Why it matters:** a Delta commit is the creation of `_delta_log/<version>.json`; two writers must never both create one version
- **Any other answer: stop.** A store or proxy that drops the header cannot guarantee safe concurrent commits; then only one writer per table (the scripts' per-table locks give that if nothing else writes)
- **No DynamoDB lock table:** deltalake 1.x commits with conditional PUT

The same guarantee, used three times

- **Delta commits:** `lake publish`, `lake maintain` (deltalake 1.x)
- **Iceberg commits:** drishti-lakeio's conditional PUT, probed at start (a store without it: single writer)
- **Content store on S3:** create `If-None-Match: *`, update `If-Match: <ETag>`; a lost race is `409 DRS-5057`

Step 5: storage tiers by business date


```
deploy/aws/delta-lake/setup-lifecycle.sh --print | head -20     # the rules, no AWS call
deploy/aws/delta-lake/setup-lifecycle.sh
```

You should see `setup-lifecycle.sh: 129 rules[1] for acme-drishti-prod-lake` (three tables from 2019).

- **By date, not by age:** age is the time since an object was written, wrong for a restated or backfilled day; one rule per table per business month, with transition dates
- Standard → Standard-IA when the month's last day is 90 days old (`TIER_IA_AFTER_DAYS`); → Glacier Instant Retrieval at two years (`TIER_GLACIER_IR_AFTER_DAYS`)
- Never Glacier Flexible Retrieval or Deep Archive (hours to restore); never an S3 expiry of current lake objects
- `_delta_log/` never matches: commits and checkpoints stay in Standard; objects under 128 KB stay where they are
- Rules exist three months ahead; maintain.sh rewrites them nightly; S3 allows 1,000 rules and the script refuses more
- **Risk tables:** 7 years kept, old months in Glacier IR, still read in milliseconds (official numbers stay reproducible)

[1] 3 tables × (5 whole years 2019–2023 already in Glacier IR + 37 months 2024-01 to 2027-01, three months ahead) + 3 base rules = 129

Step 6: EFS access points, one per node per path


```
deploy/aws/ecs/setup-efs.sh --dry-run
deploy/aws/ecs/setup-efs.sh
```

You should see one access point id per line (`fsap-0a1b…`) and `setup-efs.sh: 14 access points on fs-… for three nodes`.

Owner	Access points	Root (identical path inside the container)
each node (drishti-a, -b, -c)	data, config, sutras	`/drishti/<env>/<node>/opt/drishti/<path>` → `/opt/drishti/<path>`
consoles	data, config, sutras	`/drishti/<env>/consoles/opt/drishti/<path>`
loader	data	`/drishti/<env>/loader/opt/drishti/data` (scratch lakes, table locks)
maintenance	data	`/drishti/<env>/maintenance/opt/drishti/data`

- Every access point runs as uid/gid 10001 (the image's user); 3 × 3 + 3 + 1 + 1 = 14
- Idempotent: the access point's name is its client token, so running again returns the same ids
- The rule from AWS_DEPLOYMENT.md: every path a container writes maps to the identical path on EFS

Step 7: Cloud Map seeds and the Drishti*net transport


```
deploy/aws/ecs/setup-cloud-map.sh        # once for the namespace (it takes a minute) …
deploy/aws/ecs/setup-cloud-map.sh        # … and again for one service per node
```

- ECS registers each node's task under `drishti-a.drishti.local` with the task's own address
- Servers advertise `http://<task IP>:18480`, never loopback (refused)
- A new node finds the cluster through its seed's Cloud Map name, then learns the rest by gossip
- The loader's data-load signal (`DRISHTI_SIGNAL_URL`) also uses a node's Cloud Map name

Gossip on its own port

- node.yaml `gossip-port`: auto · a number · off
- busy nodes never suspected: 12 s saturated, no suspicion

HTTP/2 between nodes

- routing.yaml `peer-protocol: http2` (h2c), one connection per peer
- 1 connection instead of 227; p99 117 ms vs 370 ms

On ECS: a fixed gossip port PROPOSED

- set `gossip-port` to a number (e.g. 18490) on every node
- open it node to node in the `drishti-node` security group

Step 8: the images


```
./mvnw -q clean package -DskipTests
docker build -f deploy/docker/ecs/server.Dockerfile  -t "$ECR_REGISTRY/drishti-server-ecs:<version>" .
docker build -f deploy/docker/ecs/console.Dockerfile -t "$ECR_REGISTRY/drishti-console-ecs:<version>" .
docker build -f deploy/docker/ecs/loader.Dockerfile  -t "$ECR_REGISTRY/drishti-loader-ecs:<version>" .  # carries env
docker push …                                         # each of the three, after aws ecr get-login-password
```

Image	Built from	Entry point checks / role
`drishti-server-ecs`	`deploy/docker/ecs/server.Dockerfile`	server-entrypoint.sh: EFS mounted and writable, real IPv4 (never loopback), first-start node.yaml and peers; health check
`drishti-console-ecs`	`deploy/docker/ecs/console.Dockerfile`	console-entrypoint.sh: the console, a separate container from the server
`drishti-loader-ecs`	`deploy/docker/ecs/loader.Dockerfile`	loader-entrypoint.sh: EFS mounted and writable; runs the lake scripts; copies deploy/aws/ with your env (no secret)

- Loader tasks read the same settings: `DRISHTI_AWS_ENV=/opt/drishti/app/deploy/aws/env`; rebuild the loader image when they change
- The loader image carries the Python lake tools (`lake build`, `lake publish`, `lake maintain`, `drishti.py convert`)
- The build context is the repository root; nothing is built or run on a developer laptop by the scripts

Step 9: task definitions, services and schedules


```
deploy/aws/ecs/register.sh --dry-run
deploy/aws/ecs/register.sh                          # --nodes --lake-jobs
deploy/aws/ecs/register.sh --consoles 4             # optional: a console service of its own
```

You should see: a task definition and a service per node (`drishti-prod-drishti-a`, …), the loader, maintenance and report task definitions, three schedules (`drishti-prod-load-day`, `-maintain`, `-report`), then `register.sh: done`.

- The node task sets `DRISHTI_DELTA_ROOT=s3://<lake bucket>/lake`: every shipped Delta connector reads the lake in S3 with no edit, **the risk lake's** `risk-store`, `exposure-store`, `var-results` and `exposure-results` included
- The official-results publishers default to the same root and write it on `s3://` (Delta, or Iceberg); the task role needs `s3:PutObject` there (slide 49)
- Templates: `deploy/aws/ecs/templates/` → node-task, node-service, console-task, console-service, lake-task, schedule-target
- `--cache nvme` (EC2 launch type) mounts the host's `/mnt/nvme/drishti/<node>` at `/opt/drishti/nvme` (slide 22)
- Schedules: `LOAD_SCHEDULE` (22:30 Mon–Fri), `MAINTAIN_SCHEDULE` (Tue–Sat 02:30), `REPORT_SCHEDULE` (daily)

Step 10: the first load


```
deploy/aws/delta-lake/load-day.sh --dry-run --date 2026-10-20 --tables trading/trade_endur
deploy/aws/delta-lake/load-day.sh --date 2026-10-20 --tables trading/trade_endur
```

You should see the steps of the daily load (slide 125), then:

```
lake publish: ok exit=0 published <n> file(s), <bytes> bytes, dates 2026-10-20 to s3://…/trade_endur version=0 (new table)
load-day.sh: business date 2026-10-20: exit 0
```

On ECS the same runs as the loader task:

```
aws ecs run-task --task-definition drishti-prod-loader \
  --overrides '{"containerOverrides":[{"name":"loader","command":["deploy/aws/delta-lake/load-day.sh","--date","2026-10-20"]}]}' …
```

In the console

`TRD <id>` on 2026-10-20 opens the trade; **Admin → Health** shows `trade-endur UP (engine: native)`; `DATAMAP trade` counts the day's trades and checks their joins against the lake.

The dumps are JSON Lines

One object a line, `.jsonl.gz`. JSON arrays still load; a CSV export is converted first: `drishti.py convert trades.csv` (CSV input is deprecated).

Step 11: alarms and the report


```
deploy/aws/delta-lake/setup-alarms.sh; deploy/aws/delta-lake/report.sh
deploy/aws/iceberg/setup-alarms.sh;    deploy/aws/iceberg/report.sh
```

You should see `setup-alarms.sh: 17 alarms put`[1] for three Delta tables (four per table, two for the bucket and EFS, one per node), and the report (slide 86).

Alarm	Fires when
`drishti-<env>-<table>-load-failed`	a load of the table failed (nothing of it was published)
`drishti-<env>-<table>-not-loaded`	no successful load for three days running (a weekend is two)
`drishti-<env>-<table>-load-slow`	a load took longer than `ALARM_PUBLISH_DEADLINE_HOURS`
`drishti-<env>-<table>-log-long` **(Delta)**	readers replay more than `ALARM_COMMITS_SINCE_CHECKPOINT` commits
snapshots (Iceberg)	more than `ALARM_ICEBERG_SNAPSHOTS` (500) snapshots kept
`drishti-<env>-lake-s3-5xx`	more than `ALARM_S3_5XX_PER_5MIN` server errors in 5 minutes, twice in 15
`drishti-<env>-efs-io-limit`	EFS PercentIOLimit above 80 %
`drishti-<env>-<node>-down`	a node's service has no running task for 5 minutes

All notify `ALARM_SNS_TOPIC_ARN`; `report.sh` is scheduled daily (`REPORT_SCHEDULE`) and sends its numbers to CloudWatch (Iceberg: `IcebergSnapshots`, `NewestBusinessDate`). Heap, Drishti*net member dead and backup failed come with the platform's agent; Drishti*net health is on **Admin → Health** and **Admin → Net**.

[1] 4 per table × 3 tables + 2 (bucket, EFS) + 1 per node × 3 = 17

IAM, KMS and the network for Iceberg

Role	Delta lake (built policies)	Iceberg lake (`iceberg/iam/*.json`)
server task	GET, LIST on `lake/`; `kms:Decrypt`	the same on `iceberg/`; with Glue: `glue:GetCatalog`, `GetDatabase`, `GetTable` (read)
loader task	PUT data files and commits; **never delete** (explicit deny)	PUT data, manifests, `vN.metadata.json`, `version-hint.text`; the deny can stay (old metadata removal is best effort); no `--keep-days` in the loader; with Glue: `glue:UpdateTable`, `CreateTable`
maintenance task	retention, checkpoints, vacuum: the only role that deletes	`--keep-days`, `--expire-hours`, `--relayout`: deletes data, manifests, old metadata
KMS	SSE-KMS bucket default, S3 Bucket Keys	unchanged: the native writer sends no SSE header; bucket default applies. Iceberg table encryption: refused

Network

- S3 gateway endpoint, bucket policy allowing only it (as Delta): unchanged
- Path-based tables need nothing else
- Glue catalog: an interface endpoint for Glue, or traffic to the Glue endpoint through the proxy settings
- Vended credentials: STS endpoint (already among the Delta prerequisites)

The proxy and TLS settings carry over: `proxy.*` and `tls.*` apply to the REST catalog's client and to the object store (ICEBERG_CONNECTOR.md: Security; Outbound proxy).
Private CA for the catalog's own HTTPS: import into the JVM truststore.

Columns of every table

Column	Type	Content
`id`	string	the entity's id (the pack's key: `TRADE_ID`, `VECTOR_ID`, …)
`doc`	string	the whole JSON document as the source sent it (a vector replaced by a marker)
`business_date`	date	partition
partition columns	string	`PRODUCT_TYPE`
promoted columns	string or double	copies of the fields searches, lists, links and BI read: `layout.<kind>.columns`
vector columns	`list<double>` (or `list<float>`)	`layout.<kind>.vectors`: `PNL`, `SCENARIO_PV`, `MTM`; one column, the length is data

- Choosing what to promote is a pack decision (DELTA_CONNECTOR.md §4); `layout.<kind>.type-ahead` adds promoted names to the type-ahead (`USEQ Ford` finds `F`)
- Adding a promoted column changes the table's columns: `lake publish` refuses a day whose columns differ → runbook (slide 127); in place: planned, not built
- Vectors are never stored twice: the document keeps `{"$vector": "PNL"}` and a fetched document gets the vector back

Columns and table properties

Column (field id)	Type	Content
`id` (1)	string, required	the entity's id: `MX-20000001`
`doc` (2)	string	the whole JSON document, exactly what a view shows
`business_date` (3)	date, required	the identity partition
promoted (`TRADE_ID`, `PORTFOLIO`, …)	string / double	the pack's `layout.<kind>.columns`; `LEGS.CURRENCY` → `LEGS__CURRENCY`

Property	Value	Why
partition spec / sort order	`identity(business_date)` / `id ASC`	a day is a unit; engines that honour the sort order keep it
`format-version`	2	position and equality deletes
`write.parquet.row-group-size-bytes`	1,048,576	one read decodes about 3 MB of documents (sized compressed)
`write.parquet.compression-codec`	zstd	about 0.7 KB per 6.5 KB trade (measured)
`write.metadata.metrics.column.doc` / `.id`	none / full	no bounds of documents; exact id bounds per file
`write.metadata.previous-versions-max`	50 (delete after commit)	old metadata files removed
`commit.retry.num-retries`	10	concurrent day commits retry
`drishti.layout.file-rows`	250,000	rows per file; maintenance re-cuts drifted days

What of the Iceberg specification is covered

Area	Covered	Refused or later
Table metadata	every v1/v2 field; unknown fields kept when it writes the next version	format version 3 (refused)
Types	boolean, int, long, float, double, decimal, date, time, timestamp(tz), string, uuid, fixed, binary, struct, list, map	v3 types: timestamp_ns, variant, geometry
Schema evolution	by field id: add, drop, rename, reorder, int→long, float→double, decimal widening; name mapping	—
Partitioning	identity, bucket (Murmur3), truncate, year, month, day, hour, void; several specs (evolution)	—
Manifests	lists and manifests (Avro), data and delete manifests, inherited ids and sequence numbers, bounds	—
Deletes	position and equality deletes, by the sequence-number rules	v3 deletion vectors (Puffin): later
Snapshots	current snapshot, snapshot log (time travel by *known at*), the `main` ref	named branches and tags: later
Data files, encryption	Parquet	ORC, Avro data files; encrypted tables (refused)
Catalogs	path-based, a metadata file named directly, REST	Hive metastore, JDBC (only behind a REST catalog)

Maintenance as built: IcebergMaintenance

Option	What it does (one Iceberg commit each)
`--keep-days N`	deletes business days older than the newest N on or before `--as-of`: metadata only; > 50 % of rows needs `--force-drop`
`--relayout` (`--force`)	rewrites only days that drifted (overlapping id ranges, oversized files, delete files, missing columns), sorted, validated against the snapshot read
`--columns kind:path,…`	adds promoted columns and rewrites every day, filling them from the documents
`--rewrite-manifests`	regroups manifests, one per month of business dates
`--expire-hours H`	expires snapshots older than H (168), keeps the current, deletes files only they referenced
`--remove-orphans --older-than-hours`	files no snapshot references (a load that died before its commit), 72 h at least, with a dry run
Admin → Lakes, `lake maintain`	an Iceberg table maintained as a Delta one: ownership rule, lock, schedules, history, a Format column
The 7-year floor of the risk domains	`--floor-days 2557` for `risk` and `risk-results`: no date dropped, no snapshot expired inside it

Never apply S3 lifecycle **expiry** to a table's files, in either format: they delete files the metadata still names.

The Delta daily load in detail, and its runbook


```
EventBridge Scheduler (LOAD_SCHEDULE, LOAD_SCHEDULE_ZONE)
  └─► ECS RunTask drishti-<env>-loader ─► loader-entrypoint.sh (EFS mounted? writable?) ─► load-day.sh
        for each table of LAKE_TABLES_FILE:
          flock  /opt/drishti/data/lake-locks/<domain>--<table>.lock   (held: exit 3)
          aws s3 sync   s3://landing/<date>/<source>/  →  scratch/in/<table>/      (JSON Lines)
          drishti.py lake build   --from scratch/in/<table> -r --kind trade --pack config/packs/trading
                                  --mode overwrite-dates --batch-rows 250000 --partition-by PRODUCT_TYPE …
          aws s3 sync   scratch/lake/trading/trade_endur/ → s3://lake/…/trade_endur/  --exclude "_delta_log/*"
          drishti.py lake publish --from scratch/lake/… --to s3://…/trade_endur --dates <date> [--signal …]
          CloudWatch Drishti/Lake: LoadSucceeded, PublishedBusinessDate, LoadSeconds
        scratch removed (always)
```

Exit	Meaning	What to do
0	every table published	check: CloudWatch Logs `/drishti/<env>`, stream prefix loader, `LoadSucceeded` 1 per table; `report.sh --tables <table>`: newest is today; `TRD <id>` on today's date; `DATAMAP` counts the day
2	published with rejected documents (the summary says which)	the build summary names the reasons; fix the source and restate the day
1	a table failed: its S3 table is unchanged	`lake build failed` or `lake publish failed` says why; fix, then `load-day.sh --date <d> --tables <t>`
3	a table was locked	maintenance or a backfill held the table's lock; the next run loads it, or run it by hand after

The date defaults to today in `LOAD_SCHEDULE_ZONE`: an evening load in New York is already the next day in UTC.

A backfill and a restatement

A backfill

1. `backfill.sh --dry-run --from … --to … --tables <t>`: check the dates and commands
2. Run it as a loader task per table (`aws ecs run-task … --overrides` with `backfill.sh …`)
3. It stops at a failing date with the resume command; fix and resume
4. `setup-lifecycle.sh` (or the next maintain.sh) moves the backfilled years to their tiers

A restatement

1. The source re-sends the JSON Lines dumps under the landing prefix of the day
2. `backfill.sh --dates <d>` (whole day) or `load-day.sh --date <d> --tables <t> --mode overwrite-partitions`
3. Readers switch at the commit; the data-load signal (if set) refreshes the servers at once

A slow table

Symptom	Cause	Fix
first read of a day slow, then fast	id map not yet on disk; footers cold	normal; `warm-dates` (3) warms the newest days at start
every read of a day slow	the day is not sorted by id, or row groups are large	restate the day after fixing the order (slide 17)
searches slow or partial	a field the search uses is not promoted	promote it; publish then refuses the next day: backfill into `trade_endur_v2` and switch `table.trade` (in place: not built)
the snapshot refresh slow (Health)	many live files or a long log (intraday commits)	report.sh: behind and file counts; run maintain.sh; split the table
503 Slow Down	one prefix overloaded	partition by `PRODUCT_TYPE` (or finer); S3 scales a prefix within minutes
searches read documents on one table	deletion vectors in its files (written by Spark)	compaction rewrites the files without them

A restated risk date

1. The risk batch rewrites the whole date (a new `BATCH_ID`, `RESTATEMENT` + 1) as JSON Lines
2. Load it like any table: `load-day.sh --date <d> --tables risk/pnl-vector` (overwrite-dates)
3. The data-load signal reloads the date: a new hot file (new inputs' digest); the intraday layer is dropped
4. The official capture runs again: the next `RESULT_VERSION`, appended; the earlier version is kept
5. Check: `drishti.py var official` lists both versions; BI shows the latest; `publishError` empty

If the date did not load

`DRS-5203` (two runs or scenario sets on one date): the engine answers the date before, with a warning. Fix the batch, rewrite the date whole.

Nothing is lost

Time travel keeps the old inputs 7 years; the official table keeps every result version; a capture keeps its vector. "What did we report?" and "what is the corrected number?" both stay answerable.

An upgrade: the connector files follow the pack

A file nobody edited

- replaced in full by the pack's new template; the old one kept as `.before-upgrade-….bak`; audited `connector-template-upgrade`

A file you edited

- gains every key the template has, at any depth; your values stay; lists keep their order

A value that differs

- listed in Admin → Connectors → Template updates, with **Use the pack's value** per key

the stamp is in the file's text: `# drishti-template: pack=… template=…`

Tested on the real files

`ConnectorUpgradeTest` starts a server on an earlier release's generated files, an edited `risk-store` and an edited older `var-whatif`: unedited files equal the new templates, edits stay, VaR, what-if and exposure answer.

On ECS

Deploy the new images; at start each node brings its connector files (EFS or the content store) in step. Check **Template updates** after the first start.

Data written before a new column (the limit letters' `LIMIT_LETTER`) skips that hierarchy alone with a `DRS-5211` warning; MGMT, LEGAL and exposure answer as before (`HierarchyUpgradeWithoutLettersTest`).

Runbooks: symptoms, causes, fixes

Symptom	Likely cause	Fix
`DOWN: cannot reach …`	root, bucket or catalog namespace not reachable	check `root` / `uri`, the task role, `s3.*`
`DOWN: no Iceberg tables under …`	nothing loaded in that domain, or the wrong `domain` / `namespace`	load it; check the domain
new data not visible	the refresh has not run yet	wait `refresh-seconds` (10) or send the data-load signal
single reads slow, `daysWithDeletes` **> 0**	another engine's equality deletes	`IcebergMaintenance --relayout`
single reads slow on another engine's table	large row groups or unsorted days	`--relayout --force`, or that engine's sorted rewrite
searches say `partial: true`	a field the query reads is not promoted	add it to the layout, then `--columns`
***known at* finds nothing**	the snapshots of that time were expired	keep more history (`--expire-hours`)
log: `… does not honour conditional writes … single-writer`	the store ignores `If-None-Match`	one writer per table, or a REST catalog
loader: `iceberg: the load failed: <why>`	the first reason in the failure's chain	fix and load the day again (idempotent)

deploy/ and tools/lake/: the files


```
deploy/
├── docker/              compose, server and console Dockerfiles
│   └── ecs/             server, console, loader images + entry points
├── aws/
│   ├── env.example      every setting; copy to deploy/aws/env
│   ├── lib/common.sh    --help, --dry-run, --env, templates
│   ├── delta-lake/      setup-buckets, setup-iam,
│   │                    check-conditional-writes, setup-lifecycle,
│   │                    setup-alarms, load-day, backfill, maintain,
│   │                    report, lake-maintenance.s3.yaml, iam/
│   ├── iceberg/         the same set for Iceberg
│   └── ecs/             setup-efs, setup-cloud-map, register,
│                        templates/*.json
├── grafana/             the Grafana dashboard
└── lake-maintenance.yaml   lake maintain's local example
```

```
tools/lake/
├── build.py  publish.py  maintain.py
├── demo.py   stage.py  writers.py
├── stores.py  stores_equality.py
├── s3sim.py   javaload.py (--engine java)
└── test_publish  test_maintain
    test_s3_simulators  test_vectors
    test_retention  test_names
tools/samplegen/load_rows.py
tools/convertcmd.py   drishti.py convert
```

`drishti.py lake publish` (TOOLS.md §4.11) and the S3 tests (§4.12) run in the drill; `tools/test_deploy_aws.py` runs every script with --help and --dry-run.

Conventions every script follows

- bash with `set -euo pipefail` and the licence header
- AWS CLI v2 and nothing else; Python only through `tools/drishti.py`
- `--help` on every script; `--dry-run`: each changing call printed as `+ aws …`, each read as `? aws …`, none made
- Settings from one file: `deploy/aws/env` (from env.example; placeholders only; git-ignored), or `--env <file>`
- Safe to run again (idempotent): buckets, policies, access points and rules are kept and their settings applied again
- A placeholder left stops the script and names it
- Locks: load, backfill and maintenance take the table's flock on EFS; a held lock exits 3
- Data in is JSON Lines; a CSV source is converted first with `drishti.py convert`
- Tested: `tools/test_deploy_aws.py` (every script with --help and --dry-run against a fake settings file and an aws that fails if called; every template rendered; lifecycle rules; env git-ignored); `test_publish.py` and the S3 simulator tests run in the drill

How the numbers were derived (1/4)

Slide		How it was arrived at
2: Executive summary	1	10 TB × USD 23 + 46 TB × 12.5 + 144 TB × 4 a TB-month = USD 1,381; the tier split by business date and the S3 list prices (one Region) from AWS_DELTA_LAKE.md 12.3; EFS: 200,000 GB × USD 0.30 a GB-month (EFS Standard list price) = USD 60,000
2: Executive summary	2	2,300,000 positions × 509 scenarios × 8 B = 9.37 GB a date (float64; × 4 B = 4.68 GB float32); positions: the product owner's bank scale; 509: the scenario set (VAR_GUIDE.md). 9.37 GB × 0.75 (Parquet against raw, VAR_GUIDE.md 27.4: 'about 70–80 % of raw') = 7.02 GB a date; × 252 business dates = 1.77 TB a year; × 1,764 dates (7 years) = 12.4 TB; ESTIMATED
6: Why S3 and not EFS: the comparison	1	10 TB × USD 23 + 46 TB × 12.5 + 144 TB × 4 a TB-month = USD 1,381; the tier split by business date and the S3 list prices (one Region) from AWS_DELTA_LAKE.md 12.3
6: Why S3 and not EFS: the comparison	4	AWS_DELTA_LAKE.md §2's figure for EFS Infrequent Access; it implies USD 0.016 a GB-month over 200,000 GB; to be confirmed against the current EFS price list
52: Exposure paths: the large table, and what is kept for 7 years	1	trades × P paths × G grid dates × 4 B: 1,000,000 × 2,000 × 50 × 4 = 400 GB; × 5,000 × 100 = 2.0 TB; float64 doubles them; 100,000 sets × P × G × 4 B for the V matrices
52: Exposure paths: the large table, and what is kept for 7 years	2	2.0 TB × 1,764 dates = 3.53 PB; kept tiers (EXPOSURE_GUIDE.md 27.3): 32 dates × 2.0 TB = 64 TB + 50 TB (a year of set matrices) + 17 TB (84 month-ends) + 0.85 TB (statistics, contributions, captures) = 132 TB; the guide says about 150 TB

How the numbers were derived (2/4)

Slide		How it was arrived at
60: What 2.3 million positions put in the lake each day	1	2,300,000 positions × 509 scenarios × 8 B = 9.37 GB a date (float64; × 4 B = 4.68 GB float32); positions: the product owner's bank scale; 509: the scenario set (VAR_GUIDE.md)
60: What 2.3 million positions put in the lake each day	2	9.37 GB × 0.75 (Parquet against raw, VAR_GUIDE.md 27.4: 'about 70–80 % of raw') = 7.02 GB a date; × 252 business dates = 1.77 TB a year; × 1,764 dates (7 years) = 12.4 TB; ESTIMATED
60: What 2.3 million positions put in the lake each day	3	7.02 GB ÷ 0.512 GB = 13.7; ÷ 0.25 GB = 28.1
60: What 2.3 million positions put in the lake each day	4	7.02 GB ÷ 1 MB = 7,024 ranged GETs
60: What 2.3 million positions put in the lake each day	5	15,000 × 509 × 8 B = 61.1 MB a date a currency
63: Reading one VaR date at bank scale: lake, hot file, memory	1	the guide's 10–40 s for a pass over about 4 GB of Parquet × 7.02 ÷ 4 = 18–70 s (VAR_GUIDE.md 27.4); ESTIMATED

How the numbers were derived (3/4)

Slide		How it was arrived at
63: Reading one VaR date at bank scale: lake, hot file, memory	2	× 23 = 2,300,000 ÷ 100,000: linear from VarTierBenchmark, measured at 100,000 trades × 5 dates on one workstation (VAR_GUIDE.md 27.3): 237.7 MB × 23 = 5.47 GB; × 60 disk-dates = 328 GB
63: Reading one VaR date at bank scale: lake, hot file, memory	3	80 MB × 23 = 1.84 GB; whole bank 4.68 GB float32 ÷ 60 GB/s (measured rate) = 78 ms; ÷ 3 GB/s NVMe (assumed, the VaR deck's figure) = 1.6 s
91: Monthly cost: Delta, and what changes with Iceberg	1	10 TB × USD 23 + 46 TB × 12.5 + 144 TB × 4 a TB-month = USD 1,381; the tier split by business date and the S3 list prices (one Region) from AWS_DELTA_LAKE.md 12.3
91: Monthly cost: Delta, and what changes with Iceberg	2	250 reads/s × 2 GETs × 2,592,000 s = 1.30 billion GETs × USD 0.40 a million = USD 518, plus loads and lists
91: Monthly cost: Delta, and what changes with Iceberg	3	3 tasks × (6 vCPU × USD 0.04048 + 16 GB × USD 0.004445) × 730 h = USD 688 (Fargate list prices, us-east-1, Linux/x86; the Region is an assumption)
96: Three machines: sizing a machine, 200K to 10M positions	1	Heap H = 0.323 + 2 × (1,023 B × P + 19.5 KB × 20,000 nodes) (fitted at 200,000; the 1,000,000 run adds P × N ÷ 16 for the nodes' bit sets, which the planned compressed sets remove: BENCHMARK.md 4.8; 20,000 nodes = 5,000 hierarchy nodes + 15,000 portfolios: ASSUMPTION); at 2M: 0.323 + 2 × (2.05 + 0.39) = 5.20 GB; -Xmx need (H + 0.13) ÷ 0.61 = 8.73 GB. Hot file 2,783 B × P; a member holds 2/3; RAM = heap need + two open dates' page cache + a load (float64 spill 4,072 B × P × 2/3 + its hot file) + 7 GB, ÷ 0.61 up to a size = 28.9 → 48 GB. NVMe: 60 dates + a load, ÷ 0.61. Full width: the capacity model (RESOURCES_AND_SCALE 9.5) × P ÷ 2.3M, page cache, load and NVMe × 2/3; at 2M RAM 97.3 GB. Lake 1,362 B × P × 252 (measured, synthetic) or 3,054 B (design 0.75 × raw). EXTRAPOLATED from the 200,000-position fit, with the memory work done.

How the numbers were derived (4/4)

Slide		How it was arrived at
99: Three machines on AWS: the monthly cost	1	Canada (Central), on demand, 730 h: m7i.8xlarge = USD 3.5952 ÷ 2 = 1.7976 an hour × 3 × 730 = 3,937; gp3 3 × (1,000 GB × 0.088 + 3,000 IOPS × 0.0055 + 375 MiB/s × 0.044) = 363; S3 2,000 GB × 0.025 + 10 = 60; ALB 0.02475 × 730 + 2 LCU × 0.0088 × 730 = 31; zones 1,000 GB × 2 × 0.01 = 20; Fargate 60 h × (8 × 0.04456 + 32 × 0.004865) = 31; NAT and endpoints 173, CloudWatch 217, secrets 4 as the VaR deck's model; subtotal 4,836 + RDS 600–1,300 (ASSUMPTION: db.m7g.xlarge–2xlarge Multi-AZ, us-east-1 list + ~10%) = 5,436–6,136. Savings Plans: EC2 −26.9% / −49.5%, Fargate −20.0% / −45.0% (1 / 3 years, no upfront) = −1,065 / −1,964. Graviton: −19% of the EC2 line (ASSUMPTION, us-east-1 list).

Glossary

Term	Meaning
Delta Lake	a table format: Parquet data files plus `_delta_log/`, an ordered log of JSON commits and Parquet checkpoints
Apache Iceberg	a table format: Parquet data files plus `metadata/`: a metadata JSON, a manifest list per snapshot, manifests
Snapshot / version	the table as one commit left it; time travel reads an earlier one until expiry or vacuum
Catalog	where an engine finds a table's current metadata: path-based (`version-hint.text`) or a REST service (Glue, Polaris)
Conditional PUT	an S3 create with `If-None-Match: *`: a second create of the same key is refused (412); commits rely on it
Known at	a view as the data stood at an instant (`X-Drishti-Known-At`, `knownAt`), read through time travel
Promoted column	a field copied out of the document into a column so searches, lists and BI never read documents
Vector column	`list<double>`: a trade's P&L over the scenarios, its scenario PVs, or an exposure path, in one column

Term	Meaning
Id map	the per-day map from an entity's id to its file and row group; on disk, memory-mapped
Hot file	a VaR or exposure date in memory-mapped float32 form on local disk, rebuilt from the lake
Landing bucket	where the sources drop their JSON Lines dumps, by date and source, for 30 days
lake build / publish / maintain	the lake tools: JSON Lines to a table; a built day committed into S3; retention, checkpoints, vacuum, relayout
IcebergLoader / IcebergMaintenance	Drishti's Java loader and maintenance for Iceberg tables, in the server's exec jar
Retention floor	the 7 years the `risk` and `risk-results` domains always keep, whatever the settings say
Drishti*net	several Drishti servers as one: gossip, routing, replicas and shards, HTTP/2 between nodes
Data-load signal	`POST /api/v1/packs/{pack}/loads`: tells the servers a date has landed; they refresh and check it

Related documents

Document	What it holds
docs/architecture/AWS_DELTA_LAKE.md	the Delta lake on AWS: every section, command and number of the scripts
docs/architecture/AWS_DEPLOYMENT.md · SUPER_SCALE.md · SCALE_STRATEGY.md	ECS, EFS, the identical-path rule, DR; §22: Iceberg on AWS and official results to S3; S1–S8
docs/architecture/DRISHTI_ICEBERG.md	the native Iceberg engine: protocol coverage, catalogs, commits, two connectors
docs/connectors/ICEBERG_CONNECTOR.md · ICEBERG_EXTERNAL_CONNECTOR.md	the `iceberg` connector, loader, maintenance; `iceberg-external`
docs/connectors/DELTA_CONNECTOR.md · CONNECTOR_FILES.md	how Drishti reads Delta; layout, vectors, history; connector upgrades
docs/admin/LAKE_MAINTENANCE.md · docs/guides/TOOLS.md	lake maintain, Admin → Lakes, the 7-year floor; lake build, publish, the S3 tests
docs/admin/NET.md · DRISHTI_NET.md · CONTENT_STORE.md	Drishti*net: gossip port, HTTP/2, replicas and shards; the content store on S3
docs/guides/DATA_FORMATS.md	JSON Lines is the standard; CSV deprecated; official results as JSONL
docs/guides/VAR_GUIDE.md · EXPOSURE_GUIDE.md	the risk lake: inputs, vectors, intraday, official results, sizes
docs/architecture/VAR_SHARDING.md · PARALLEL_LOADER.md	VaR sharded across servers, Fargate drain; the parallel Java loader
docs/connectors/DEMO_DATA.md · docs/guides/PUBLIC_DATA.md	the 85-table demo lake, mixed storage; the public-data packs
Drishti-VaR-Engine.pptx · Drishti-Design-and-Evidence.pptx	the companion decks: the VaR engine; the whole product

Drishti दृष्टि

Thank you

See what your data hasn't told you yet.

ANY DATA. ANY DOMAIN. ONE GRAMMAR.

Drishti on AWS: the lake, Delta or Iceberg

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